

Part I: Lorentzian Einstein-Klein-Gordon

This section serves as a minimal test of the xAct pipeline: we derive the Einstein-Klein-Gordon field equations, project them to spherical-polar components, and verify against the Schwarzschild vacuum solution. The physics here is standard; the purpose is to confirm that the variational, chart, and component machinery is functioning correctly before proceeding to the Euclidean sector.

Scalar field and Klein-Gordon Lagrangian

The kinetic scalar X is defined as minus one half the squared gradient of the scalar field.

$$-\frac{1}{2} (\bar{\mathcal{D}}_\alpha \phi) (\bar{\mathcal{D}}^\alpha \phi)$$

The Lagrangian density for Einstein gravity with a free massless scalar field:

$$\frac{R[\bar{\mathcal{D}}]}{2\kappa} - \frac{1}{2} (\bar{\mathcal{D}}_\alpha \phi) (\bar{\mathcal{D}}^\alpha \phi)$$

Covariant field equations via VarD

The metric field equation is obtained by varying the Lagrangian with respect to the inverse metric:

$$\frac{R[\bar{\mathcal{D}}]_{\alpha\beta}}{2\kappa} - \frac{1}{2} (\bar{\mathcal{D}}_\alpha \phi) (\bar{\mathcal{D}}_\beta \phi) == 0$$

The scalar field equation is obtained by varying the Lagrangian with respect to the scalar field:

$$\bar{\mathcal{D}}_\alpha \bar{\mathcal{D}}^\alpha \phi == 0$$

Chart and metric components

Lorentzian line element with signature $(-,+,+,+)$:

$$\begin{array}{cccc} -f[r] & 0 & 0 & 0 \\ 0 & h[r] & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin[\theta]^2 \end{array}$$

Component field equations

Metric equation components

$(t,t):$

$$\frac{-\frac{f'[r]^2}{f[r]} + f'[r] \left(\frac{4}{r} - \frac{h'[r]}{h[r]} \right) + 2 f''[r]}{8 \kappa h[r]} == 0$$

$(t,r):$

True

$(t,\theta):$

True

$(t,\phi):$

True

$(r,r):$

$$\frac{1}{8} (-4 \phi'[r]^2 + \frac{f[r] (4 f[r] + r f'[r]) h'[r] + r h[r] (f'[r]^2 - 2 f[r] f''[r])}{\kappa r f[r]^2 h[r]}) == 0$$

$(r,\theta):$

True

$(r,\phi):$

True

$(\theta,\theta):$

$$\frac{2 - \frac{r f'[r]}{h[r]} + \frac{r h'[r]}{h[r]^2}}{4 \kappa} == 0$$

$(\theta,\phi):$

True

$(\phi,\phi):$

$$\frac{\sin[\theta]^2 (-r h[r] f'[r] + f[r] (-2 h[r] + 2 h[r]^2 + r h'[r]))}{4 \kappa f[r] h[r]^2} == 0$$

Scalar equation components

$$\frac{r h[r] f'[r] \phi'[r] + f[r] (-r h'[r] \phi'[r] + 2 h[r] (2 \phi'[r] + r \phi''[r]))}{2 r f[r] h[r]^2} == 0$$

Verification: Schwarzschild solution (vacuum)

We substitute the Schwarzschild metric functions and set the scalar field to zero, then verify that all field equations vanish identically.

All field equations vanish identically for the Schwarzschild vacuum with trivial scalar. Verified.

Part II: Euclidean Riemannian gravity (Mukohyama-Uzan)

This section contains the core scientific computation. We work in Euclidean (Riemannian) signature following Mukohyama-Uzan (1301.1361), deriving the field equations for the Horndeski G4 sector with a Lemaitre line element (cf. Eq. 4.2 of 2208.02943). The metric and scalar field equations are projected to spherical-polar components and exported as a standalone equation set for further analysis.

Euclidean manifold and metric

We define a Riemannian manifold with signature $(+, +, +, +)$.

Riemannian gravity Lagrangian

The Lagrangian combines the Horndeski G4 sector (Eq. 3.6 of 1301.1361) with the quadratic-curvature invariants of the ELST theory (Eq. 4 of 2310.17266).

The kinetic scalar of the clock field:

$$(\bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^a \phi)$$

The Horndeski G4 sector consists of the Ricci scalar coupled to G4, the shift-symmetric potential K, and the Galileon combination involving second covariant derivatives of the clock field.

The quadratic-curvature sector adds three invariants: the square of the Ricci scalar, the contraction of the Ricci tensor with itself, and the Kretschner scalar (the full contraction of the Riemann tensor), with constant couplings c_1, c_2, c_3 .

$$\begin{aligned} & \mathcal{K}[(\bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^a \phi)] + c_2 R[\bar{\mathcal{D}}]_{ab} R[\bar{\mathcal{D}}]^{ab} + \mathcal{G}_4[(\bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^a \phi)] R[\bar{\mathcal{D}}] + c_1 R[\bar{\mathcal{D}}]^2 + \\ & c_3 R[\bar{\mathcal{D}}]_{abcd} R[\bar{\mathcal{D}}]^{abcd} - 2 (-(\bar{\mathcal{D}}_c \bar{\mathcal{D}}_d \phi) (\bar{\mathcal{D}}^c \bar{\mathcal{D}}^d \phi) + (\bar{\mathcal{D}}_c \bar{\mathcal{D}}^c \phi) (\bar{\mathcal{D}}_d \bar{\mathcal{D}}^d \phi)) \mathcal{G}_4'[(\bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^a \phi)] \end{aligned}$$

Euclidean field equations via VarD

Metric field equation from varying the gravitational action with respect to the inverse Euclidean metric:

Metric field equation (before introducing shorthand tensors):

[illegible]

Scalar field equation from varying the gravitational action with respect to the clock field:

Scalar field equation (covariant):

$$\begin{aligned}
& -2 R[\bar{\mathcal{D}}](\bar{\mathcal{D}}_a \bar{\mathcal{D}}^a \phi) \mathcal{G}_4'[\mathcal{X}_\varepsilon] + 4 (\bar{\mathcal{D}}_a \bar{\mathcal{D}}_b \bar{\mathcal{D}}^b \bar{\mathcal{D}}^a \phi) \mathcal{G}_4'[\mathcal{X}_\varepsilon] - 2 (\bar{\mathcal{D}}_a R[\bar{\mathcal{D}}]) (\bar{\mathcal{D}}^a \phi) \mathcal{G}_4'[\mathcal{X}_\varepsilon] - \\
& 4 (\bar{\mathcal{D}}_b \bar{\mathcal{D}}^b \bar{\mathcal{D}}_a \bar{\mathcal{D}}^a \phi) \mathcal{G}_4'[\mathcal{X}_\varepsilon] - 2 (\bar{\mathcal{D}}_a \bar{\mathcal{D}}^a \phi) \mathcal{K}'[\mathcal{X}_\varepsilon] + 8 (\bar{\mathcal{D}}_a \bar{\mathcal{D}}_c \bar{\mathcal{D}}^c \phi) (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}^b \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] - \\
& 4 R[\bar{\mathcal{D}}] (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^b \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] - 16 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}_c \bar{\mathcal{D}}^c \phi) (\bar{\mathcal{D}}^b \bar{\mathcal{D}}_a \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] + \\
& 8 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}^b \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}_b \bar{\mathcal{D}}^c \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] + 4 (\bar{\mathcal{D}}_a \bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}^b \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}^c \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] - \\
& 8 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}^b \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}^c \bar{\mathcal{D}}_a \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] + 8 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}^b \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}^c \bar{\mathcal{D}}_b \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] + \\
& 8 (\bar{\mathcal{D}}^b \bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}_b \phi) (\bar{\mathcal{D}}^c \bar{\mathcal{D}}_a \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] - 8 (\bar{\mathcal{D}}_a \bar{\mathcal{D}}_c \bar{\mathcal{D}}_b \phi) (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}^c \bar{\mathcal{D}}^b \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] + \\
& 8 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}_c \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^c \bar{\mathcal{D}}^b \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] - 12 (\bar{\mathcal{D}}_a \bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}_b \phi) (\bar{\mathcal{D}}^c \bar{\mathcal{D}}^b \phi) \mathcal{G}_4''[\mathcal{X}_\varepsilon] - \\
& 4 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^b \phi) \mathcal{K}''[\mathcal{X}_\varepsilon] + 8 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^b \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}^c \phi) (\bar{\mathcal{D}}_d \bar{\mathcal{D}}^d \phi) \mathcal{G}_4^{(3)}[\mathcal{X}_\varepsilon] - \\
& 16 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}^b \phi) (\bar{\mathcal{D}}_c \bar{\mathcal{D}}_b \phi) (\bar{\mathcal{D}}^c \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}_d \bar{\mathcal{D}}^d \phi) \mathcal{G}_4^{(3)}[\mathcal{X}_\varepsilon] + \\
& 16 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}^b \phi) (\bar{\mathcal{D}}^c \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}_d \bar{\mathcal{D}}_c \phi) (\bar{\mathcal{D}}^d \bar{\mathcal{D}}_b \phi) \mathcal{G}_4^{(3)}[\mathcal{X}_\varepsilon] - \\
& 8 (\bar{\mathcal{D}}^a \phi) (\bar{\mathcal{D}}_b \bar{\mathcal{D}}_a \phi) (\bar{\mathcal{D}}^b \phi) (\bar{\mathcal{D}}_d \bar{\mathcal{D}}_c \phi) (\bar{\mathcal{D}}^d \bar{\mathcal{D}}^c \phi) \mathcal{G}_4^{(3)}[\mathcal{X}_\varepsilon] = 0
\end{aligned}$$

Metric field equation with shorthand tensors:

$$\begin{aligned}
& \mathcal{G}_4[X_\varepsilon] R[\bar{\mathcal{D}}]_{ab} + 2 c_2 R[\bar{\mathcal{D}}]_a{}^c R[\bar{\mathcal{D}}]_{bc} + 2 c_1 R[\bar{\mathcal{D}}]_{ab} R[\bar{\mathcal{D}}] + 2 c_3 R[\bar{\mathcal{D}}]_a{}^{cde} R[\bar{\mathcal{D}}]_{bcde} - \\
& c_1 (\bar{\mathcal{D}}_a \bar{\mathcal{D}}_b R[\bar{\mathcal{D}}]) - c_1 (\bar{\mathcal{D}}_b \bar{\mathcal{D}}_a R[\bar{\mathcal{D}}]) - c_2 (\bar{\mathcal{D}}_c \bar{\mathcal{D}}_a R[\bar{\mathcal{D}}]_b{}^c) - c_2 (\bar{\mathcal{D}}_c \bar{\mathcal{D}}_b R[\bar{\mathcal{D}}]_a{}^c) + \\
& c_2 (\bar{\mathcal{D}}_c \bar{\mathcal{D}}^c R[\bar{\mathcal{D}}]_{ab}) + 2 c_1 \bar{g}_{ab} (\bar{\mathcal{D}}_c \bar{\mathcal{D}}^c R[\bar{\mathcal{D}}]) + 2 c_3 (\bar{\mathcal{D}}_c \bar{\mathcal{D}}_d R[\bar{\mathcal{D}}]_a{}^c{}_b{}^d) + c_2 \bar{g}_{ab} (\bar{\mathcal{D}}_d \bar{\mathcal{D}}_c R[\bar{\mathcal{D}}]^{cd}) + \\
& 2 c_3 (\bar{\mathcal{D}}_d \bar{\mathcal{D}}_c R[\bar{\mathcal{D}}]_a{}^c{}_b{}^d) - \nabla \nabla \phi_b{}^c \nabla \nabla \phi_{ca} \mathcal{G}_4'[X_\varepsilon] - \nabla \nabla \phi_a{}^c \nabla \nabla \phi_{cb} \mathcal{G}_4'[X_\varepsilon] + \nabla \nabla \phi_{ab} \nabla \nabla \phi_c{}^c \mathcal{G}_4'[X_\varepsilon] + \\
& \nabla \nabla \phi_{ba} \nabla \nabla \phi_c{}^c \mathcal{G}_4'[X_\varepsilon] - \nabla \phi^c \nabla \nabla \nabla \phi_{abc} \mathcal{G}_4'[X_\varepsilon] + 2 \nabla \phi_b \nabla \nabla \nabla \phi_a{}^c \mathcal{G}_4'[X_\varepsilon] - \nabla \phi^c \nabla \nabla \nabla \phi_{bac} \mathcal{G}_4'[X_\varepsilon] + \\
& 2 \nabla \phi_a \nabla \nabla \nabla \phi_b{}^c \mathcal{G}_4'[X_\varepsilon] + \nabla \phi^c \nabla \nabla \nabla \phi_{cab} \mathcal{G}_4'[X_\varepsilon] + \nabla \phi^c \nabla \nabla \nabla \phi_{cba} \mathcal{G}_4'[X_\varepsilon] - \nabla \phi_b \nabla \nabla \nabla \phi_{ac} \mathcal{G}_4'[X_\varepsilon] - \\
& \nabla \phi_a \nabla \nabla \nabla \phi_{bc} \mathcal{G}_4'[X_\varepsilon] - \nabla \phi_b \nabla \nabla \nabla \phi_{ca} \mathcal{G}_4'[X_\varepsilon] - \nabla \phi_a \nabla \nabla \nabla \phi_{cb} \mathcal{G}_4'[X_\varepsilon] + 2 \nabla \nabla \phi_{cd} \nabla \nabla \phi^{cd} \bar{g}_{ab} \mathcal{G}_4'[X_\varepsilon] - \\
& 2 \nabla \nabla \phi_c{}^c \nabla \nabla \phi_d{}^d \bar{g}_{ab} \mathcal{G}_4'[X_\varepsilon] - 2 \nabla \phi^c \nabla \nabla \nabla \phi_c{}^d \bar{g}_{ab} \mathcal{G}_4'[X_\varepsilon] + 2 \nabla \phi^c \nabla \nabla \nabla \phi_{dc} \bar{g}_{ab} \mathcal{G}_4'[X_\varepsilon] + \\
& \nabla \phi_a \nabla \phi_b R[\bar{\mathcal{D}}] \mathcal{G}_4'[X_\varepsilon] + \nabla \phi_a \nabla \phi_b \mathcal{K}'[X_\varepsilon] - 4 \nabla \phi^c \nabla \phi^d \nabla \nabla \phi_{ac} \nabla \nabla \phi_{bd} \mathcal{G}_4''[X_\varepsilon] + \\
& 2 \nabla \phi^c \nabla \phi^d \nabla \nabla \phi_{ab} \nabla \nabla \phi_{cd} \mathcal{G}_4''[X_\varepsilon] + 2 \nabla \phi^c \nabla \phi^d \nabla \nabla \phi_{ba} \nabla \nabla \phi_{cd} \mathcal{G}_4''[X_\varepsilon] + \\
& 2 \nabla \phi_a \nabla \phi_b \nabla \nabla \phi_{cd} \nabla \nabla \phi^{cd} \mathcal{G}_4''[X_\varepsilon] - 2 \nabla \phi_b \nabla \phi^c \nabla \nabla \phi_a{}^d \nabla \nabla \phi_{dc} \mathcal{G}_4''[X_\varepsilon] - \\
& 2 \nabla \phi_a \nabla \phi^c \nabla \nabla \phi_b{}^d \nabla \nabla \phi_{dc} \mathcal{G}_4''[X_\varepsilon] - 2 \nabla \phi_b \nabla \phi^c \nabla \nabla \phi_{dc} \nabla \nabla \phi_a{}^d \mathcal{G}_4''[X_\varepsilon] - \\
& 2 \nabla \phi_a \nabla \phi^c \nabla \nabla \phi_{dc} \nabla \nabla \phi_b{}^d \mathcal{G}_4''[X_\varepsilon] + 4 \nabla \phi_b \nabla \phi^c \nabla \nabla \phi_{ac} \nabla \nabla \phi_d{}^d \mathcal{G}_4''[X_\varepsilon] + \\
& 4 \nabla \phi_a \nabla \phi^c \nabla \nabla \phi_{bc} \nabla \nabla \phi_d{}^d \mathcal{G}_4''[X_\varepsilon] - 2 \nabla \phi_a \nabla \phi_b \nabla \nabla \phi_c{}^c \nabla \nabla \phi_d{}^d \mathcal{G}_4''[X_\varepsilon] + \\
& 4 \nabla \phi^c \nabla \phi^d \nabla \nabla \phi_{ed} \nabla \nabla \phi_c{}^e \bar{g}_{ab} \mathcal{G}_4''[X_\varepsilon] - 4 \nabla \phi^c \nabla \phi^d \nabla \nabla \phi_{cd} \nabla \nabla \phi_e{}^e \bar{g}_{ab} \mathcal{G}_4''[X_\varepsilon] == 0
\end{aligned}$$

Scalar field equation with shorthand tensors:

$$\begin{aligned}
& -4 \nabla \nabla \nabla \phi_a{}^b{}_b \mathcal{G}_4'[X_\varepsilon] + 4 \nabla \nabla \nabla \phi^{ab}{}_{ba} \mathcal{G}_4'[X_\varepsilon] - 2 \nabla \nabla \phi_a{}^a R[\bar{\mathcal{D}}] \mathcal{G}_4'[X_\varepsilon] - \\
& 2 \nabla \phi^a (\bar{\mathcal{D}}_a R[\bar{\mathcal{D}}]) \mathcal{G}_4'[X_\varepsilon] - 2 \nabla \nabla \phi_a{}^a \mathcal{K}'[X_\varepsilon] + 8 \nabla \nabla \phi_a{}^c \nabla \nabla \phi^{ab} \nabla \nabla \phi_{bc} \mathcal{G}_4''[X_\varepsilon] - \\
& 12 \nabla \nabla \phi_a{}^a \nabla \nabla \phi_{bc} \nabla \nabla \phi^{bc} \mathcal{G}_4''[X_\varepsilon] + 4 \nabla \nabla \phi_a{}^a \nabla \nabla \phi_b{}^b \nabla \nabla \phi_c{}^c \mathcal{G}_4''[X_\varepsilon] - \\
& 8 \nabla \phi^a \nabla \nabla \phi^{bc} \nabla \nabla \phi_{abc} \mathcal{G}_4''[X_\varepsilon] + 8 \nabla \phi^a \nabla \nabla \phi_b{}^b \nabla \nabla \phi_a{}^c \mathcal{G}_4''[X_\varepsilon] - 16 \nabla \phi^a \nabla \nabla \phi_b{}^a \nabla \nabla \phi_{bc} \mathcal{G}_4''[X_\varepsilon] + \\
& 8 \nabla \phi^a \nabla \nabla \phi^{bc} \nabla \nabla \phi_{cba} \mathcal{G}_4''[X_\varepsilon] + 8 \nabla \phi^a \nabla \nabla \phi_b{}^a \nabla \nabla \phi_{bc} \mathcal{G}_4''[X_\varepsilon] - 8 \nabla \phi^a \nabla \nabla \phi_b{}^b \nabla \nabla \phi_{ca} \mathcal{G}_4''[X_\varepsilon] + \\
& 8 \nabla \phi^a \nabla \nabla \phi_b{}^a \nabla \nabla \phi_{cb} \mathcal{G}_4''[X_\varepsilon] - 4 \nabla \phi^a \nabla \phi^b \nabla \nabla \phi_{ab} R[\bar{\mathcal{D}}] \mathcal{G}_4''[X_\varepsilon] - 4 \nabla \phi^a \nabla \phi^b \nabla \nabla \phi_{ab} \mathcal{K}''[X_\varepsilon] - \\
& 8 \nabla \phi^a \nabla \phi^b \nabla \nabla \phi_{ab} \nabla \nabla \phi_{cd} \nabla \nabla \phi^{cd} \mathcal{G}_4^{(3)}[X_\varepsilon] + 16 \nabla \phi^a \nabla \phi^b \nabla \nabla \phi_c{}^d \nabla \nabla \phi_a{}^c \nabla \nabla \phi_{db} \mathcal{G}_4^{(3)}[X_\varepsilon] - \\
& 16 \nabla \phi^a \nabla \phi^b \nabla \nabla \phi_{cb} \nabla \nabla \phi_c{}^a \nabla \nabla \phi_d{}^d \mathcal{G}_4^{(3)}[X_\varepsilon] + 8 \nabla \phi^a \nabla \phi^b \nabla \nabla \phi_{ab} \nabla \nabla \phi_c{}^c \nabla \nabla \phi_d{}^d \mathcal{G}_4^{(3)}[X_\varepsilon] == 0
\end{aligned}$$

Euclidean chart and pre-computed components

Euclidean Lemaitre line element with signature (+,+,+,+):

$$\begin{array}{cccc}
1 & 0 & 0 & 0 \\
0 & 1 - \mathcal{A}[\tau, \rho] & 0 & 0 \\
0 & 0 & r[\tau, \rho]^2 & 0 \\
0 & 0 & 0 & r[\tau, \rho]^2 \sin[\theta]^2
\end{array}$$

Component field equations (term-by-term)

Euclidean metric field equations (component form):

(τ, ρ) :

$$\begin{aligned}
& \frac{1}{8r[\tau, \rho]^4 (-1 + \mathcal{A}[\tau, \rho])^5} \\
& (-16(4c_1 + c_2)(-1 + \mathcal{A}[\tau, \rho])^3 r^{(0,1)}[\tau, \rho]^4 - 24(4c_1 + c_2)r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])^2 r^{(0,1)}[\tau, \rho]^3 \mathcal{A}^{(0,1)}[\tau, \rho] + \\
& 16(8c_1 + 3c_2 + 4c_3)(-1 + \mathcal{A}[\tau, \rho])^5 r^{(1,0)}[\tau, \rho]^2 (-1 + r^{(1,0)}[\tau, \rho]^2) + \\
& 8(4c_1 + c_2)r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])^4 (r^{(0,2)}[\tau, \rho](2 - 6r^{(1,0)}[\tau, \rho]^2) - r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 3r^{(1,0)}[\tau, \rho]^3 \mathcal{A}^{(1,0)}[\tau, \rho] - 2(-1 + \mathcal{A}[\tau, \rho])r^{(2,0)}[\tau, \rho] + 2(-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]) - \\
& 4(-1 + \mathcal{A}[\tau, \rho])r^{(0,1)}[\tau, \rho]^2 (-4(4c_1 + c_2 + 2(2c_1 + c_2 + 2c_3))r^{(1,0)}[\tau, \rho]^2) - \\
& 2(4c_1 + c_2)r[\tau, \rho](6r^{(0,2)}[\tau, \rho] - 3r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 2r^{(2,0)}[\tau, \rho]) + \\
& 4\mathcal{A}[\tau, \rho]^3 (4c_1 + c_2 + 2(2c_1 + c_2 + 2c_3))r^{(1,0)}[\tau, \rho]^2 - r[\tau, \rho]^2 \mathcal{G}_4'[X_\delta] \phi^{(1,0)}[\tau, \rho]^2 + \\
& (4c_1 + c_2)r[\tau, \rho]r^{(2,0)}[\tau, \rho] + r[\tau, \rho]^2 (4(4c_1 + c_2) \mathcal{A}^{(0,1)}[\tau, \rho]^2 + \\
& 8c_1 \mathcal{A}^{(0,2)}[\tau, \rho] + 2c_2 \mathcal{A}^{(0,2)}[\tau, \rho] + 8\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 + \\
& 4\mathcal{G}_4'[X_\delta] (2\phi^{(0,1)}[\tau, \rho]^2 + \phi^{(1,0)}[\tau, \rho]^2) - 2c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 5c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 12c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 4c_1 \mathcal{A}^{(2,0)}[\tau, \rho] - 2c_2 \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_3 \mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& 2\mathcal{A}[\tau, \rho]^2 (-6(4c_1 + c_2 + 2(2c_1 + c_2 + 2c_3))r^{(1,0)}[\tau, \rho]^2) - \\
& 3(4c_1 + c_2)r[\tau, \rho](2r^{(0,2)}[\tau, \rho] - r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 2r^{(2,0)}[\tau, \rho]) + \\
& r[\tau, \rho]^2 (4\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 + \\
& \mathcal{G}_4'[X_\delta] (4\phi^{(0,1)}[\tau, \rho]^2 + 6\phi^{(1,0)}[\tau, \rho]^2) + (2c_1 - c_2 - 4c_3) \mathcal{A}^{(2,0)}[\tau, \rho])) + \\
& \mathcal{A}[\tau, \rho] (12(4c_1 + c_2 + 2(2c_1 + c_2 + 2c_3))r^{(1,0)}[\tau, \rho]^2 + 12(4c_1 + c_2)r[\tau, \rho] \\
& (2r^{(0,2)}[\tau, \rho] - r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + r^{(2,0)}[\tau, \rho]) + \\
& r[\tau, \rho]^2 (-2(4c_1 + c_2) \mathcal{A}^{(0,2)}[\tau, \rho] - 16\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 - \\
& 4\mathcal{G}_4'[X_\delta] (4\phi^{(0,1)}[\tau, \rho]^2 + 3\phi^{(1,0)}[\tau, \rho]^2) + 2c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 5c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 12c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 8c_1 \mathcal{A}^{(2,0)}[\tau, \rho] + 4c_2 \mathcal{A}^{(2,0)}[\tau, \rho] + 16c_3 \mathcal{A}^{(2,0)}[\tau, \rho])) + \\
& 4r[\tau, \rho]r^{(0,1)}[\tau, \rho] (14(4c_1 + c_2)r[\tau, \rho]^2 \mathcal{A}^{(0,1)}[\tau, \rho]^3 + (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(0,1)}[\tau, \rho] \\
& (2(4c_1 + c_2)(-1 + 3r^{(1,0)}[\tau, \rho]^2) - 2r[\tau, \rho](3(4c_1 + c_2)r^{(0,2)}[\tau, \rho] + \\
& (20c_1 + 9c_2 + 16c_3)r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - (12c_1 + 3c_2 + 4c_3)r^{(2,0)}[\tau, \rho]) + \\
& \mathcal{A}[\tau, \rho]^2 (-2(4c_1 + c_2) + 6(4c_1 + c_2)r^{(1,0)}[\tau, \rho]^2 - 4r[\tau, \rho]^2 \mathcal{G}_4'[X_\delta] \phi^{(1,0)}[\tau, \rho]^2 + \\
& 2(12c_1 + 3c_2 + 4c_3)r[\tau, \rho]r^{(2,0)}[\tau, \rho]) + r[\tau, \rho]^2 (-52c_1 \mathcal{A}^{(0,2)}[\tau, \rho] - \\
& 13c_2 \mathcal{A}^{(0,2)}[\tau, \rho] - 8\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 (\phi^{(0,1)}[\tau, \rho]^2 + \phi^{(1,0)}[\tau, \rho]^2) - \\
& 4\mathcal{G}_4'[X_\delta] (3\phi^{(0,1)}[\tau, \rho]^2 + \phi^{(1,0)}[\tau, \rho]^2) + 14c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 8c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 18c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 4c_1 \mathcal{A}^{(2,0)}[\tau, \rho] + 6c_2 \mathcal{A}^{(2,0)}[\tau, \rho] + 20c_3 \mathcal{A}^{(2,0)}[\tau, \rho]) - \\
& 2\mathcal{A}[\tau, \rho] (2(4c_1 + c_2)(-1 + 3r^{(1,0)}[\tau, \rho]^2) - r[\tau, \rho](3(4c_1 + c_2)r^{(0,2)}[\tau, \rho] + \\
& (20c_1 + 9c_2 + 16c_3)r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2(12c_1 + 3c_2 + 4c_3) \\
& r^{(2,0)}[\tau, \rho]) + r[\tau, \rho]^2 (-4\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 - 2\mathcal{G}_4'[X_\delta] \\
& (3\phi^{(0,1)}[\tau, \rho]^2 + 2\phi^{(1,0)}[\tau, \rho]^2) + (2c_1 + 3c_2 + 10c_3) \mathcal{A}^{(2,0)}[\tau, \rho])) + \\
& r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])^2 (-2(-1 + \mathcal{A}[\tau, \rho]) (2(4c_1 + c_2)r^{(0,3)}[\tau, \rho] - 8(-1 + \mathcal{A}[\tau, \rho]) \mathcal{G}_4'[X_\delta] \\
& \phi^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + 8\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 - \\
& 8\mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 - 4c_1 \mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,1)}[\tau, \rho] - \\
& 5c_2 \mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,1)}[\tau, \rho] - 8c_3 \mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,1)}[\tau, \rho] + 8c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + \\
& 4c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + 8c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - 2c_2 r^{(2,1)}[\tau, \rho] - \\
& 8c_3 r^{(2,1)}[\tau, \rho] + 2c_2 \mathcal{A}[\tau, \rho]r^{(2,1)}[\tau, \rho] + 8c_3 \mathcal{A}[\tau, \rho]r^{(2,1)}[\tau, \rho]) + \\
& r[\tau, \rho] (8c_1 \mathcal{A}^{(0,3)}[\tau, \rho] + 2c_2 \mathcal{A}^{(0,3)}[\tau, \rho] - 8\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho]^2 - \\
& (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho]^2) (-2\phi^{(0,2)}[\tau, \rho] + \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - \\
& 12c_1 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - 7c_2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - \\
& 16c_3 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + 8(-1 + \mathcal{A}[\tau, \rho]) \mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho]
\end{aligned}$$

$$\begin{aligned}
& (-2\phi^{(0,2)}[\tau, \rho] + \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) \phi^{(2,0)}[\tau, \rho]) - \\
& 8c_1 \mathcal{A}^{(2,1)}[\tau, \rho] - 6c_2 \mathcal{A}^{(2,1)}[\tau, \rho] - 16c_3 \mathcal{A}^{(2,1)}[\tau, \rho] + \\
& 8c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 6c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 16c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho])) - \\
& 4r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho])^3 (4(-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{G}_4'[X_\varepsilon] (-1 + 3r^{(1,0)}[\tau, \rho]^2) \phi^{(1,0)}[\tau, \rho]^2 + \\
& 8(-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{G}_4''[X_\varepsilon] r^{(1,0)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^4 + \\
& 48c_1 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 20c_2 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 32c_3 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 2c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 18c_1 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 7c_2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 8c_3 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 24c_1 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + \\
& 10c_2 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + \\
& 16c_3 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + \\
& 16c_1 r^{(1,1)}[\tau, \rho]^2 - \\
& 16c_1 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho]^2 + \\
& 48c_1 r^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] + \\
& 20c_2 r^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] + \\
& 32c_3 r^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] - \\
& 48c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] - \\
& 20c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] - \\
& 32c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] - \\
& 48c_1 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 12c_2 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 16c_3 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 48c_1 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 12c_2 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 16c_3 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 4c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 8c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 4c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 8c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 32c_1 r^{(2,0)}[\tau, \rho]^2 - 12c_2 r^{(2,0)}[\tau, \rho]^2 - \\
& 16c_3 r^{(2,0)}[\tau, \rho]^2 + 64c_1 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho]^2 + \\
& 24c_2 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho]^2 + 32c_3 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho]^2 - \\
& 32c_1 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]^2 - 12c_2 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]^2 - \\
& 16c_3 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]^2 - 4c_1 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 4c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 12c_1 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 4c_2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 12c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 4c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 32c_1 r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 12c_2 r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + \\
& 16c_3 r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - 64c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 24c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 16c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] -
\end{aligned}$$

[illegible]

$$\begin{aligned}
& (6\mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,0)}[\tau, \rho] + 5r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + 2(-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho])) + \\
& (-1 + \mathcal{A}[\tau, \rho])(-16c_1r^{(0,4)}[\tau, \rho] - 4c_2r^{(0,4)}[\tau, \rho] + 16c_1\mathcal{A}[\tau, \rho]r^{(0,4)}[\tau, \rho] + \\
& 4c_2\mathcal{A}[\tau, \rho]r^{(0,4)}[\tau, \rho] - 16\mathcal{G}_4''[X_\varepsilon]\phi^{(0,1)}[\tau, \rho]^2\phi^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho] + \\
& 16\mathcal{A}[\tau, \rho]\mathcal{G}_4''[X_\varepsilon]\phi^{(0,1)}[\tau, \rho]^2\phi^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho] - \\
& 16\mathcal{G}_4''[X_\varepsilon]\phi^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]^3 + 32\mathcal{A}[\tau, \rho]\mathcal{G}_4''[X_\varepsilon]\phi^{(0,2)}[\tau, \rho] \\
& r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]^3 - 16\mathcal{A}[\tau, \rho]^2\mathcal{G}_4''[X_\varepsilon]\phi^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]^3 + \\
& 8c_1\mathcal{A}^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + 2c_2\mathcal{A}^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 8\mathcal{G}_4''[X_\varepsilon]\phi^{(0,1)}[\tau, \rho]^2r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho] - 8\mathcal{A}[\tau, \rho]\mathcal{G}_4''[X_\varepsilon]\phi^{(0,1)}[\tau, \rho]^2 \\
& r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho] + 8\mathcal{G}_4''[X_\varepsilon]r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]^4\mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 16\mathcal{A}[\tau, \rho]\mathcal{G}_4''[X_\varepsilon]r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]^4\mathcal{A}^{(1,0)}[\tau, \rho] + 8\mathcal{A}[\tau, \rho]^2\mathcal{G}_4''[X_\varepsilon]r^{(1,0)}[\tau, \rho] \\
& \phi^{(1,0)}[\tau, \rho]^4\mathcal{A}^{(1,0)}[\tau, \rho] + 6c_1r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^3 + c_2r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^3 - \\
& 2c_3r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^3 + 16c_1r^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] + 4c_2r^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] - \\
& 16c_1\mathcal{A}[\tau, \rho]r^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] - 4c_2\mathcal{A}[\tau, \rho]r^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] + \\
& 16c_1\mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,2)}[\tau, \rho] + 4c_2\mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,2)}[\tau, \rho] - 16c_1\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] \\
& r^{(1,2)}[\tau, \rho] - 4c_2\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,2)}[\tau, \rho] + 8c_1r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,2)}[\tau, \rho] + \\
& 2c_2r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,2)}[\tau, \rho] - 8c_1\mathcal{A}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,2)}[\tau, \rho] - \\
& 2c_2\mathcal{A}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,2)}[\tau, \rho] - 4\mathcal{G}_4[X_\varepsilon]r^{(2,0)}[\tau, \rho] + 12\mathcal{G}_4[X_\varepsilon]\mathcal{A}[\tau, \rho]r^{(2,0)}[\tau, \rho] - \\
& 12\mathcal{G}_4[X_\varepsilon]\mathcal{A}[\tau, \rho]^2r^{(2,0)}[\tau, \rho] + 4\mathcal{G}_4[X_\varepsilon]\mathcal{A}[\tau, \rho]^3r^{(2,0)}[\tau, \rho] - \\
& 12c_1\mathcal{A}^{(1,0)}[\tau, \rho]^2r^{(2,0)}[\tau, \rho] - 3c_2\mathcal{A}^{(1,0)}[\tau, \rho]^2r^{(2,0)}[\tau, \rho] + \\
& 12c_1\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^2r^{(2,0)}[\tau, \rho] + 3c_2\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^2r^{(2,0)}[\tau, \rho] + \\
& 16c_1r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] + 3c_2r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 4c_3r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] - 16c_1\mathcal{A}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 3c_2\mathcal{A}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] + 4c_3\mathcal{A}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] \\
& \mathcal{A}^{(2,0)}[\tau, \rho] - 16c_1r^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] - 4c_2r^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 32c_1\mathcal{A}[\tau, \rho]r^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] + 8c_2\mathcal{A}[\tau, \rho]r^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 16c_1\mathcal{A}[\tau, \rho]^2r^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] - 4c_2\mathcal{A}[\tau, \rho]^2r^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] + \\
& r^{(0,2)}[\tau, \rho](-8(4c_1 + c_2)\mathcal{A}^{(0,2)}[\tau, \rho] + 2(2c_1 + c_2 + 2c_3)\mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 8(c_1 - c_3)(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) - 16c_1r^{(2,2)}[\tau, \rho] - 4c_2r^{(2,2)}[\tau, \rho] + \\
& 32c_1\mathcal{A}[\tau, \rho]r^{(2,2)}[\tau, \rho] + 8c_2\mathcal{A}[\tau, \rho]r^{(2,2)}[\tau, \rho] - 16c_1\mathcal{A}[\tau, \rho]^2r^{(2,2)}[\tau, \rho] - \\
& 4c_2\mathcal{A}[\tau, \rho]^2r^{(2,2)}[\tau, \rho] + 8c_1\mathcal{A}^{(1,0)}[\tau, \rho]r^{(3,0)}[\tau, \rho] + 2c_2\mathcal{A}^{(1,0)}[\tau, \rho]r^{(3,0)}[\tau, \rho] - \\
& 16c_1\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]r^{(3,0)}[\tau, \rho] - 4c_2\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]r^{(3,0)}[\tau, \rho] + \\
& 8c_1\mathcal{A}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]r^{(3,0)}[\tau, \rho] + 2c_2\mathcal{A}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]r^{(3,0)}[\tau, \rho] + \\
& 8c_1r^{(1,0)}[\tau, \rho]\mathcal{A}^{(3,0)}[\tau, \rho] + 2c_2r^{(1,0)}[\tau, \rho]\mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 16c_1\mathcal{A}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(3,0)}[\tau, \rho] - 4c_2\mathcal{A}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 8c_1\mathcal{A}[\tau, \rho]^2r^{(1,0)}[\tau, \rho]\mathcal{A}^{(3,0)}[\tau, \rho] + 2c_2\mathcal{A}[\tau, \rho]^2r^{(1,0)}[\tau, \rho]\mathcal{A}^{(3,0)}[\tau, \rho])) = 0
\end{aligned}$$

$(\tau, \rho):$

$$\begin{aligned}
& \frac{1}{2r[\tau, \rho]^4(-1 + \mathcal{A}[\tau, \rho])^4}(-4(-1 + \mathcal{A}[\tau, \rho])^2r^{(0,1)}[\tau, \rho]^3 \\
& (4(3c_1 + c_2 + c_3)(-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho] + (4c_1 + c_2)r[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]) + \\
& r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])r^{(0,1)}[\tau, \rho]^2 \\
& (r[\tau, \rho](4(-1 + \mathcal{A}[\tau, \rho])^2\mathcal{G}_4''[X_\varepsilon]\phi^{(0,1)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho] - 8(-1 + \mathcal{A}[\tau, \rho])\mathcal{G}_4''[X_\varepsilon]\phi^{(0,1)}[\tau, \rho]^3 \\
& \phi^{(1,0)}[\tau, \rho] - (8c_1 + 3c_2 + 4c_3)\mathcal{A}^{(0,1)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]) + \\
& 8(4c_1 + c_2)(-1 + \mathcal{A}[\tau, \rho])^2r^{(1,1)}[\tau, \rho]) + r^{(0,1)}[\tau, \rho] \\
& (16(3c_1 + c_2 + c_3)(-1 + \mathcal{A}[\tau, \rho])^4r^{(1,0)}[\tau, \rho]^3 + 4(4c_1 + c_2)r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])^3r^{(1,0)}[\tau, \rho]^2 \\
& \mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho](-16(3c_1 + c_2 + c_3) + 16(3c_1 + c_2 + c_3)\mathcal{A}[\tau, \rho]^3 - \\
& r[\tau, \rho]^2(8(2c_1 + c_2 + 2c_3)\mathcal{A}^{(0,1)}[\tau, \rho]^2 + 4(2c_1 + c_2 + 2c_3)\mathcal{A}^{(0,2)}[\tau, \rho] +
\end{aligned}$$

$$\begin{aligned}
& 16 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 - 4 c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 8 c_1 \mathcal{A}^{(2,0)}[\tau, \rho] + 8 c_3 \mathcal{A}^{(2,0)}[\tau, \rho] + 8 \mathcal{A}[\tau, \rho]^2 (-6 (3 c_1 + c_2 + c_3) + \\
& r[\tau, \rho]^2 (-2 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 + (c_1 - c_3) \mathcal{A}^{(2,0)}[\tau, \rho])) + \\
& \mathcal{A}[\tau, \rho] (48 (3 c_1 + c_2 + c_3) + r[\tau, \rho]^2 (4 (2 c_1 + c_2 + 2 c_3) \mathcal{A}^{(0,2)}[\tau, \rho] + 32 \\
& \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 - 4 c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - c_2 \\
& \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 16 c_1 \mathcal{A}^{(2,0)}[\tau, \rho] + 16 c_3 \mathcal{A}^{(2,0)}[\tau, \rho])) + \\
& r[\tau, \rho]^2 (2 (-1 + \mathcal{A}[\tau, \rho])^2 ((8 c_1 + 3 c_2 + 4 c_3) r^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& (8 c_1 + 3 c_2 + 4 c_3) \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) ((8 c_1 + 3 c_2 + 4 c_3) \\
& \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 4 (2 c_1 + c_2 + 2 c_3) (-1 + \mathcal{A}[\tau, \rho]) r^{(3,0)}[\tau, \rho])) + \\
& r[\tau, \rho] (2 \mathcal{G}_4[X_\varepsilon] (-1 + \mathcal{A}[\tau, \rho])^3 \mathcal{A}^{(1,0)}[\tau, \rho] + 8 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^4 \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 8 \mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^4 \mathcal{A}^{(1,0)}[\tau, \rho] + 56 c_1 \mathcal{A}^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 14 c_2 \mathcal{A}^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] + 20 c_1 \mathcal{A}^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 5 c_2 \mathcal{A}^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 20 c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 5 c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 8 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 \\
& \mathcal{A}^{(1,0)}[\tau, \rho] - 16 \mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 8 \mathcal{A}[\tau, \rho]^2 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] - 2 c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^3 - \\
& 3 c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^3 - 10 c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^3 + 2 c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^3 + \\
& 3 c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^3 + 10 c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^3 + \\
& 16 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^3 \phi^{(1,1)}[\tau, \rho] - 32 \mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^3 \phi^{(1,1)}[\tau, \rho] + \\
& 16 \mathcal{A}[\tau, \rho]^2 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^3 \phi^{(1,1)}[\tau, \rho] + 16 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^2 \\
& \phi^{(1,1)}[\tau, \rho] - 48 \mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^2 \phi^{(1,1)}[\tau, \rho] + \\
& 48 \mathcal{A}[\tau, \rho]^2 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^2 \phi^{(1,1)}[\tau, \rho] - \\
& 16 \mathcal{A}[\tau, \rho]^3 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^2 \phi^{(1,1)}[\tau, \rho] + 4 (-1 + \mathcal{A}[\tau, \rho])^2 \\
& \mathcal{G}_4'[X_\varepsilon] (2 \phi^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] - 2 \\
& (-1 + \mathcal{A}[\tau, \rho]) \phi^{(0,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho]) + 32 c_1 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + \\
& 8 c_2 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - 32 c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - \\
& 8 c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + 8 c_1 \mathcal{A}^{(1,2)}[\tau, \rho] + 2 c_2 \mathcal{A}^{(1,2)}[\tau, \rho] - \\
& 16 c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,2)}[\tau, \rho] - 4 c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,2)}[\tau, \rho] + 8 c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,2)}[\tau, \rho] + \\
& 2 c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,2)}[\tau, \rho] - 4 c_1 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 6 c_2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 20 c_3 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 8 c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 12 c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 40 c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 4 c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 6 c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 20 c_3 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 2 c_2 \mathcal{A}^{(3,0)}[\tau, \rho] - 8 c_3 \mathcal{A}^{(3,0)}[\tau, \rho] + 6 c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 24 c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 6 c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(3,0)}[\tau, \rho] - 24 c_3 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 2 c_2 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(3,0)}[\tau, \rho] + 8 c_3 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(3,0)}[\tau, \rho])) - \\
& r[\tau, \rho] (8 (4 c_1 + c_2) (-1 + \mathcal{A}[\tau, \rho])^4 r^{(1,0)}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho] - \\
& 2 \\
& r[\tau, \rho] \\
& (-1 + \mathcal{A}[\tau, \rho])^2 \\
& (2 (2 c_1 + c_2 + 2 c_3) \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] (-3 r^{(0,2)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho]) (4 (2 c_1 + c_2 + 2 c_3) r^{(0,3)}[\tau, \rho] r^{(1,0)}[\tau, \rho] - \\
& 2 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{G}_4'[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] (-1 + r^{(1,0)}[\tau, \rho]^2) \phi^{(1,0)}[\tau, \rho] + \\
& 4 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^3 - 4 \mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] \\
& r^{(1,0)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho]^3 - 16 c_1 r^{(0,2)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 6 c_2 r^{(0,2)}[\tau, \rho] r^{(1,1)}[\tau, \rho] -
\end{aligned}$$

$$\begin{aligned}
& 8c_3 r^{(0,2)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 4c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - \\
& 3c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 8c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - \\
& 4c_1 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,1)}[\tau, \rho] - 2c_2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,1)}[\tau, \rho] - 4c_3 r^{(1,0)}[\tau, \rho]^2 \\
& \mathcal{A}^{(1,1)}[\tau, \rho] - 16c_1 r^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 6c_2 r^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 8c_3 r^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 16c_1 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 6c_2 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 8c_3 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho]) + \\
& r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho]) (4(4c_1 + c_2) \mathcal{A}^{(0,1)}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho] + \\
& \mathcal{A}^{(0,1)}[\tau, \rho] (15(4c_1 + c_2) r^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2(4c_1 + c_2) (-1 + \mathcal{A}[\tau, \rho]) \\
& (3r^{(1,2)}[\tau, \rho] - \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho]) - r^{(1,0)}[\tau, \rho] ((20c_1 + 7c_2 + 8c_3) \\
& \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 4(2c_1 + c_2 + 2c_3) (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho])) - \\
& (-1 + \mathcal{A}[\tau, \rho]) (6(4c_1 + c_2) r^{(0,3)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 8\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^3 \\
& r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 8\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \\
& \mathcal{A}^{(1,0)}[\tau, \rho] - 8\mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 4\mathcal{G}_4[X_\delta] r^{(1,1)}[\tau, \rho] + 8\mathcal{G}_4[X_\delta] \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 4\mathcal{G}_4[X_\delta] \mathcal{A}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho] + \\
& 8c_1 \mathcal{A}^{(0,2)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 2c_2 \mathcal{A}^{(0,2)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 16c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 \\
& r^{(1,1)}[\tau, \rho] - 5c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho] - 4c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho] + \\
& 16\mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - 16\mathcal{A}[\tau, \rho] \\
& \mathcal{G}_4''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] + 16\mathcal{G}_4''[X_\delta] r^{(1,0)}[\tau, \rho] \\
& \phi^{(1,0)}[\tau, \rho]^3 \phi^{(1,1)}[\tau, \rho] - 32\mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\delta] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \phi^{(1,1)}[\tau, \rho] + \\
& 16\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4''[X_\delta] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \phi^{(1,1)}[\tau, \rho] - 4(-1 + \mathcal{A}[\tau, \rho]) \mathcal{G}_4'[X_\delta] \\
& (\phi^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 2\phi^{(0,1)}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho] - 2(-1 + \\
& \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho] (\phi^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho])) + \\
& 24c_1 r^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + 6c_2 r^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - \\
& 20c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - 7c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - \\
& 8c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + 16c_1 r^{(1,3)}[\tau, \rho] + 4c_2 r^{(1,3)}[\tau, \rho] - \\
& 16c_1 \mathcal{A}[\tau, \rho] r^{(1,3)}[\tau, \rho] - 4c_2 \mathcal{A}[\tau, \rho] r^{(1,3)}[\tau, \rho] - 8c_1 \mathcal{A}^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 2c_2 \mathcal{A}^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 8c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 2c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 16c_1 r^{(1,1)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 6c_2 r^{(1,1)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_3 r^{(1,1)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 16c_1 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 6c_2 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 8c_3 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] - 4c_2 r^{(1,0)}[\tau, \rho] \\
& \mathcal{A}^{(2,1)}[\tau, \rho] - 8c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 8c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + \\
& 4c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 8c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + \\
& 16c_1 r^{(3,1)}[\tau, \rho] + 4c_2 r^{(3,1)}[\tau, \rho] - 32c_1 \mathcal{A}[\tau, \rho] r^{(3,1)}[\tau, \rho] - \\
& 8c_2 \mathcal{A}[\tau, \rho] r^{(3,1)}[\tau, \rho] + 16c_1 \mathcal{A}[\tau, \rho]^2 r^{(3,1)}[\tau, \rho] + 4c_2 \mathcal{A}[\tau, \rho]^2 r^{(3,1)}[\tau, \rho]) + \\
& r[\tau, \rho]^3 (-2(-1 + \mathcal{A}[\tau, \rho])^4 \mathcal{K}[X_\delta] \phi^{(0,1)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + (2c_1 + c_2 + 2c_3) \\
& (\mathcal{A}^{(0,1)}[\tau, \rho] (7\mathcal{A}^{(1,0)}[\tau, \rho]^3 - 9(-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 2 \\
& (-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{A}^{(3,0)}[\tau, \rho]) - (-1 + \mathcal{A}[\tau, \rho]) (7\mathcal{A}^{(1,0)}[\tau, \rho]^2 \\
& \mathcal{A}^{(1,1)}[\tau, \rho] - 5(-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 2(-1 + \mathcal{A}[\tau, \rho]) \\
& (-2\mathcal{A}^{(1,1)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(3,1)}[\tau, \rho])))) = 0
\end{aligned}$$

$(\rho, \rho):$

$$\frac{1}{8r[\tau, \rho]^4 (-1 + \mathcal{A}[\tau, \rho])^3}$$

$$\begin{aligned}
& (-16(8c_1 + 3c_2 + 4c_3) (-1 + \mathcal{A}[\tau, \rho])^2 r^{(0,1)}[\tau, \rho]^4 + 8(4c_1 + c_2) r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho]^3 \\
& \mathcal{A}^{(0,1)}[\tau, \rho] + 16(4c_1 + c_2) (-1 + \mathcal{A}[\tau, \rho])^4 r^{(1,0)}[\tau, \rho]^2 (-1 + r^{(1,0)}[\tau, \rho]^2) +
\end{aligned}$$

$$\begin{aligned}
& 8c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 12c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 4c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 8c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 2r^{(0,2)}[\tau, \rho] \\
& \quad (2(4c_1 + c_2) r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + (12c_1 + 3c_2 + 4c_3)(-1 + \mathcal{A}[\tau, \rho]) r^{(2,0)}[\tau, \rho]) + \\
& 2c_1 \mathcal{A}^{(2,0)}[\tau, \rho] - 2c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 2c_1 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 2c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_1 r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 2c_2 r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 16c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + \\
& 4c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - 8c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 2c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + \\
& 4r[\tau, \rho]^3 (-1 + \mathcal{A}[\tau, \rho]) (-40c_1 r^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 10c_2 r^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 28c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^3 + \\
& 11c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^3 + 16c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^3 - \\
& 2\mathcal{G}_4[X_\varepsilon](-1 + \mathcal{A}[\tau, \rho])^2 (2r^{(0,2)}[\tau, \rho] - r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - \\
& 32c_1 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 8c_2 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - \\
& 16c_1 r^{(1,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - 4c_2 r^{(1,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + \\
& 16c_1 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + 4c_2 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - \\
& 32c_1 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] - 8c_2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] + \\
& 32c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] + 8c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] + \\
& 12c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] + 4c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] + \\
& 4c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] - 12c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] - \\
& 4c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] - 4c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] - \\
& 16\mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + \\
& 32\mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - \\
& 16\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - \\
& 16\mathcal{G}_4''[X_\varepsilon] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \phi^{(2,0)}[\tau, \rho] + \\
& 48\mathcal{A}[\tau, \rho] \mathcal{G}_4''[X_\varepsilon] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \phi^{(2,0)}[\tau, \rho] - \\
& 48\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4''[X_\varepsilon] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \phi^{(2,0)}[\tau, \rho] + \\
& 16\mathcal{A}[\tau, \rho]^3 \mathcal{G}_4''[X_\varepsilon] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]^3 \phi^{(2,0)}[\tau, \rho] + 4(-1 + \mathcal{A}[\tau, \rho]) \mathcal{G}_4'[X_\varepsilon] \\
& \quad (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] - 2(-1 + \mathcal{A}[\tau, \rho]) \phi^{(0,1)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] + \\
& \quad (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho] (-2\phi^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] + 2(-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho] \\
& \quad r^{(2,0)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] (\phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 4(-1 + \mathcal{A}[\tau, \rho]) \phi^{(2,0)}[\tau, \rho])) - \\
& 32c_1 r^{(0,2)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_2 r^{(0,2)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 32c_1 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 8c_2 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 48c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 20c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 32c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 48c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 20c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 32c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 8c_1 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 4c_2 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 8c_3 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 16c_1 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_2 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 16c_3 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 8c_1 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 4c_2 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + 8c_3 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 8c_1 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - 2c_2 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + \\
& 8c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + 2c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - \\
& 16c_1 r^{(2,2)}[\tau, \rho] - 4c_2 r^{(2,2)}[\tau, \rho] + 32c_1 \mathcal{A}[\tau, \rho] r^{(2,2)}[\tau, \rho] + \\
& 8c_2 \mathcal{A}[\tau, \rho] r^{(2,2)}[\tau, \rho] - 16c_1 \mathcal{A}[\tau, \rho]^2 r^{(2,2)}[\tau, \rho] - 4c_2 \mathcal{A}[\tau, \rho]^2 r^{(2,2)}[\tau, \rho] +
\end{aligned}$$

$$\begin{aligned}
& 8c_1 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 2c_2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 16c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - 4c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + \\
& 8c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 2c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + \\
& 16c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 8c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 16c_3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 32c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 16c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 32c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 16c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 8c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 16c_3 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 16c_1 r^{(4,0)}[\tau, \rho] - 4c_2 r^{(4,0)}[\tau, \rho] + \\
& 48c_1 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] + 12c_2 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] - 48c_1 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] - \\
& 12c_2 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] + 16c_1 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] + 4c_2 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] + \\
& r[\tau, \rho]^4 (8(-1 + \mathcal{A}[\tau, \rho])^3 \mathcal{K}[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 - 4\mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 4\mathcal{A}[\tau, \rho] \mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 50c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^4 - 25c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^4 - \\
& 50c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^4 - 16\mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] + \\
& 32\mathcal{A}[\tau, \rho] \mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - \\
& 16\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - 16\mathcal{G}_4'[X_\delta] \phi^{(1,1)}[\tau, \rho]^2 + \\
& 48\mathcal{A}[\tau, \rho] \mathcal{G}_4'[X_\delta] \phi^{(1,1)}[\tau, \rho]^2 - 48\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4'[X_\delta] \phi^{(1,1)}[\tau, \rho]^2 + \\
& 16\mathcal{A}[\tau, \rho]^3 \mathcal{G}_4'[X_\delta] \phi^{(1,1)}[\tau, \rho]^2 + 8\mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - \\
& 16\mathcal{A}[\tau, \rho] \mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + \\
& 8\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4'[X_\delta] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + 16\mathcal{G}_4'[X_\delta] \phi^{(0,2)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - \\
& 48\mathcal{A}[\tau, \rho] \mathcal{G}_4'[X_\delta] \phi^{(0,2)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + 48\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4'[X_\delta] \phi^{(0,2)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - \\
& 16\mathcal{A}[\tau, \rho]^3 \mathcal{G}_4'[X_\delta] \phi^{(0,2)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - 8\mathcal{G}_4'[X_\delta] \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + \\
& 24\mathcal{A}[\tau, \rho] \mathcal{G}_4'[X_\delta] \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - \\
& 24\mathcal{A}[\tau, \rho]^2 \mathcal{G}_4'[X_\delta] \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + \\
& 8\mathcal{A}[\tau, \rho]^3 \mathcal{G}_4'[X_\delta] \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] - 120c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 60c_2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - 120c_3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 120c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + 60c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 120c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - 40c_1 \mathcal{A}^{(2,0)}[\tau, \rho]^2 - \\
& 20c_2 \mathcal{A}^{(2,0)}[\tau, \rho]^2 - 40c_3 \mathcal{A}^{(2,0)}[\tau, \rho]^2 + 80c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho]^2 + \\
& 40c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho]^2 + 80c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho]^2 - 40c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho]^2 - \\
& 20c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho]^2 - 40c_3 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho]^2 - \\
& 2\mathcal{G}_4[X_\delta] (-1 + \mathcal{A}[\tau, \rho])^2 (\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2(-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) - \\
& 48c_1 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 24c_2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 48c_3 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 96c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 48c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 96c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 48c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 24c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 48c_3 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 16c_1 \mathcal{A}^{(4,0)}[\tau, \rho] - 8c_2 \mathcal{A}^{(4,0)}[\tau, \rho] - \\
& 16c_3 \mathcal{A}^{(4,0)}[\tau, \rho] + 48c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(4,0)}[\tau, \rho] + 24c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(4,0)}[\tau, \rho] + \\
& 48c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(4,0)}[\tau, \rho] - 48c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(4,0)}[\tau, \rho] - 24c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(4,0)}[\tau, \rho] - \\
& 48c_3 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(4,0)}[\tau, \rho] + 16c_1 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(4,0)}[\tau, \rho] + \\
& 8c_2 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(4,0)}[\tau, \rho] + 16c_3 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(4,0)}[\tau, \rho]) = 0
\end{aligned}$$

$(\theta, \theta):$

$$\begin{aligned}
& -((8(-1 + \mathcal{A}[\tau, \rho])^3 (1 - r^{(0,1)}[\tau, \rho]^2 - r^{(1,0)}[\tau, \rho]^2 + \mathcal{A}[\tau, \rho] (-1 + r^{(1,0)}[\tau, \rho]^2)) \\
& (-2c_1 - c_2 - 2c_3 - 2(3c_1 + c_2 + c_3) r^{(0,1)}[\tau, \rho]^2 - 6c_1 r^{(1,0)}[\tau, \rho]^2 - 2c_2 r^{(1,0)}[\tau, \rho]^2 - \\
& 2c_3 r^{(1,0)}[\tau, \rho]^2 + \mathcal{A}[\tau, \rho] (2c_1 + c_2 + 2c_3 + 2(3c_1 + c_2 + c_3) r^{(1,0)}[\tau, \rho]^2)) + \\
& 4r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho])^2 (6(3c_1 + c_2 + c_3) r^{(0,1)}[\tau, \rho]^3 \mathcal{A}^{(0,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho]
\end{aligned}$$

$$\begin{aligned}
& + r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])^2 (10c_1 + 3c_2 + 2c_3) r^{(1,0)}[\tau, \rho]^2 + 16(3c_1 + c_2 + c_3) \\
& \quad (\mathcal{A}^{(0,1)}[\tau, \rho] (10c_1 + 3c_2 + 2c_3 - 2(3c_1 + c_2 + c_3) r^{(1,0)}[\tau, \rho]^2) + 16(3c_1 + c_2 + c_3) \\
& \quad (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho]) - 2(3c_1 + c_2 + c_3) (-1 + \mathcal{A}[\tau, \rho]) \\
& \quad r^{(0,1)}[\tau, \rho]^2 (6r^{(0,2)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2(-1 + \mathcal{A}[\tau, \rho]) r^{(2,0)}[\tau, \rho]) + \\
& \quad (-1 + \mathcal{A}[\tau, \rho])^2 (2r^{(0,2)}[\tau, \rho] (-10c_1 - 3c_2 - 2c_3 + 2(3c_1 + c_2 + c_3) r^{(1,0)}[\tau, \rho]^2) + \\
& \quad (10c_1 + 3c_2 + 2c_3) r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2(3c_1 + c_2 + c_3) r^{(1,0)}[\tau, \rho]^3 \\
& \quad \mathcal{A}^{(1,0)}[\tau, \rho] + 2(10c_1 + 3c_2 + 2c_3) (-1 + \mathcal{A}[\tau, \rho]) r^{(2,0)}[\tau, \rho] - \\
& \quad 12(3c_1 + c_2 + c_3) (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]) + \\
& r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho]) (-8c_1 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(0,1)}[\tau, \rho]^2 - 3c_2 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(0,1)}[\tau, \rho]^2 - \\
& \quad 4c_3 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(0,1)}[\tau, \rho]^2 - 32c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] - \\
& \quad 12c_2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] - 16c_3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + \\
& \quad 32c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + \\
& \quad 12c_2 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + \\
& \quad 16c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] - 32c_1 r^{(0,2)}[\tau, \rho]^2 - \\
& \quad 12c_2 r^{(0,2)}[\tau, \rho]^2 - 16c_3 r^{(0,2)}[\tau, \rho]^2 + 64c_1 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho]^2 + \\
& \quad 24c_2 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho]^2 + 32c_3 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho]^2 - 32c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho]^2 - \\
& \quad 12c_2 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho]^2 - 16c_3 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho]^2 + \\
& \quad 4\mathcal{G}_4[X_6] (-1 + \mathcal{A}[\tau, \rho])^3 (1 - r^{(0,1)}[\tau, \rho]^2 - r^{(1,0)}[\tau, \rho]^2 + \mathcal{A}[\tau, \rho] (-1 + r^{(1,0)}[\tau, \rho]^2)) + \\
& \quad 8(-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{G}_4[X_6] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho])^2 + \\
& \quad 16c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 6c_2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \\
& \quad r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 8c_3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& \quad 16c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& \quad 6c_2 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& \quad 8c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& \quad 32c_1 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 12c_2 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& \quad 16c_3 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 64c_1 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& \quad 24c_2 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \\
& \quad 32c_3 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 32c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho] \\
& \quad r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 12c_2 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& \quad 16c_3 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 4c_1 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& \quad 8c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 4c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 28c_1 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& \quad 2c_2 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 8c_3 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& \quad 28c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2c_2 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& \quad 8c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 4c_1 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& \quad c_2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 4c_3 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& \quad 8c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& \quad 8c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 4c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& \quad c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 4c_3 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& \quad 64c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 32c_3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - \\
& \quad 128c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 64c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \\
& \quad \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 64c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - \\
& \quad 32c_3 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 64c_1 r^{(1,1)}[\tau, \rho]^2 - 32c_3 r^{(1,1)}[\tau, \rho]^2 - \\
& \quad 192c_1 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho]^2 + 96c_3 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho]^2 + 192c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho]^2 - \\
& \quad 96c_3 \mathcal{A}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho]^2 - 64c_1 \mathcal{A}[\tau, \rho]^3 r^{(1,1)}[\tau, \rho]^2 + 32c_3 \mathcal{A}[\tau, \rho]^3 r^{(1,1)}[\tau, \rho]^2 - \\
& \quad 64c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 12c_2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& \quad 128c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 24c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho]
\end{aligned}$$

[illegible]

$$\begin{aligned}
& 8c_2 \mathcal{A}[\tau, \rho]^4 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_3 \mathcal{A}[\tau, \rho]^4 r^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 32c_1 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + 12c_2 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + \\
& 16c_3 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - 96c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - \\
& 36c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - 48c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + \\
& 96c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + 36c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + \\
& 48c_3 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - 32c_1 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - \\
& 12c_2 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] - 16c_3 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,1)}[\tau, \rho] + \\
& 8c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 6c_2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 16c_3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] - \\
& 24c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] - 18c_2 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] - \\
& 48c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 24c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + \\
& 18c_2 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 48c_3 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] - \\
& 8c_1 \mathcal{A}[\tau, \rho]^3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] - 6c_2 \mathcal{A}[\tau, \rho]^3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] - \\
& 16c_3 \mathcal{A}[\tau, \rho]^3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(2,1)}[\tau, \rho] + 64c_1 r^{(2,2)}[\tau, \rho] + 24c_2 r^{(2,2)}[\tau, \rho] + \\
& 32c_3 r^{(2,2)}[\tau, \rho] - 256c_1 \mathcal{A}[\tau, \rho] r^{(2,2)}[\tau, \rho] - 96c_2 \mathcal{A}[\tau, \rho] r^{(2,2)}[\tau, \rho] - \\
& 128c_3 \mathcal{A}[\tau, \rho] r^{(2,2)}[\tau, \rho] + 384c_1 \mathcal{A}[\tau, \rho]^2 r^{(2,2)}[\tau, \rho] + 144c_2 \mathcal{A}[\tau, \rho]^2 r^{(2,2)}[\tau, \rho] + \\
& 192c_3 \mathcal{A}[\tau, \rho]^2 r^{(2,2)}[\tau, \rho] - 256c_1 \mathcal{A}[\tau, \rho]^3 r^{(2,2)}[\tau, \rho] - 96c_2 \mathcal{A}[\tau, \rho]^3 r^{(2,2)}[\tau, \rho] - \\
& 128c_3 \mathcal{A}[\tau, \rho]^3 r^{(2,2)}[\tau, \rho] + 64c_1 \mathcal{A}[\tau, \rho]^4 r^{(2,2)}[\tau, \rho] + 24c_2 \mathcal{A}[\tau, \rho]^4 r^{(2,2)}[\tau, \rho] + \\
& 32c_3 \mathcal{A}[\tau, \rho]^4 r^{(2,2)}[\tau, \rho] - 32c_1 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - 12c_2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 16c_3 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 128c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + \\
& 48c_2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 64c_3 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 192c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - 72c_2 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 96c_3 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 128c_1 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + \\
& 48c_2 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + 64c_3 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 32c_1 \mathcal{A}[\tau, \rho]^4 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - 12c_2 \mathcal{A}[\tau, \rho]^4 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - \\
& 16c_3 \mathcal{A}[\tau, \rho]^4 \mathcal{A}^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] - 24c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 6c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 96c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 24c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 144c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 36c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 96c_1 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 24c_2 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 24c_1 \mathcal{A}[\tau, \rho]^4 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 6c_2 \mathcal{A}[\tau, \rho]^4 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 32c_1 r^{(4,0)}[\tau, \rho] + 12c_2 r^{(4,0)}[\tau, \rho] + \\
& 16c_3 r^{(4,0)}[\tau, \rho] - 160c_1 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] - 60c_2 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] - \\
& 80c_3 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] + 320c_1 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] + 120c_2 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] + \\
& 160c_3 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] - 320c_1 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] - 120c_2 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] - \\
& 160c_3 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] + 160c_1 \mathcal{A}[\tau, \rho]^4 r^{(4,0)}[\tau, \rho] + \\
& 60c_2 \mathcal{A}[\tau, \rho]^4 r^{(4,0)}[\tau, \rho] + 80c_3 \mathcal{A}[\tau, \rho]^4 r^{(4,0)}[\tau, \rho] - 32c_1 \mathcal{A}[\tau, \rho]^5 r^{(4,0)}[\tau, \rho] - \\
& 12c_2 \mathcal{A}[\tau, \rho]^5 r^{(4,0)}[\tau, \rho] - 16c_3 \mathcal{A}[\tau, \rho]^5 r^{(4,0)}[\tau, \rho]) + \\
& r[\tau, \rho]^4 (-4(-1 + \mathcal{A}[\tau, \rho]) \mathcal{G}_4[X_\mathcal{E}] (\phi^{(0,1)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho]^2) \\
& (\phi^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2(-1 + \mathcal{A}[\tau, \rho]) \phi^{(0,1)}[\tau, \rho] \\
& (2\mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) + 2(-1 + \mathcal{A}[\tau, \rho])^2 \\
& (2\phi^{(1,1)}[\tau, \rho]^2 + (-2\phi^{(0,2)}[\tau, \rho] + \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) \phi^{(2,0)}[\tau, \rho])) + \\
& 2(-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{G}_4[X_\mathcal{E}] (-4(-1 + \mathcal{A}[\tau, \rho]) \phi^{(0,1)}[\tau, \rho] \\
& (2\mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) + \\
& \phi^{(0,1)}[\tau, \rho]^2 (3\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2(-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho]) (4(-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + 8 \\
& (-1 + \mathcal{A}[\tau, \rho]) (\phi^{(1,1)}[\tau, \rho]^2 - \phi^{(0,2)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) - \\
& \phi^{(1,0)}[\tau, \rho]^2 (\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2(-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho])) +
\end{aligned}$$

$$\begin{aligned}
& (4c_1 + c_2) (\mathcal{A}^{(0,1)}[\tau, \rho]^4 (7\mathcal{A}^{(1,0)}[\tau, \rho]^4 - 5(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) + (-1 + \mathcal{A}[\tau, \rho]) \\
& \mathcal{A}^{(0,1)}[\tau, \rho] (-9\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] + 5(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,1)}[\tau, \rho]) - \\
& (-1 + \mathcal{A}[\tau, \rho]) (5\mathcal{A}^{(1,0)}[\tau, \rho]^4 - 12(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(1,0)}[\tau, \rho]^2\mathcal{A}^{(2,0)}[\tau, \rho] + 2 \\
& \mathcal{A}^{(0,2)}[\tau, \rho](\mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) + (-1 + \mathcal{A}[\tau, \rho]) \\
& \mathcal{A}^{(1,0)}[\tau, \rho] (-2\mathcal{A}^{(1,2)}[\tau, \rho] + 5(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(3,0)}[\tau, \rho]) - 2 \\
& (-1 + \mathcal{A}[\tau, \rho]) (\mathcal{A}^{(1,1)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) (2\mathcal{A}^{(2,0)}[\tau, \rho]^2 + \mathcal{A}^{(2,2)}[\tau, \rho] - \\
& (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(4,0)}[\tau, \rho]))) / (4r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho])^5) == 0
\end{aligned}$$

(ϕ, ϕ) :

$$\begin{aligned}
& -((\sin[\theta]^2 (8(-1 + \mathcal{A}[\tau, \rho])^3 (1 - r^{(0,1)}[\tau, \rho]^2 - r^{(1,0)}[\tau, \rho]^2 + \mathcal{A}[\tau, \rho] (-1 + r^{(1,0)}[\tau, \rho]^2)) \\
& (-2c_1 - c_2 - 2c_3 - 2(3c_1 + c_2 + c_3)r^{(0,1)}[\tau, \rho]^2 - 6c_1r^{(1,0)}[\tau, \rho]^2 - 2c_2r^{(1,0)}[\tau, \rho]^2 - \\
& 2c_3r^{(1,0)}[\tau, \rho]^2 + \mathcal{A}[\tau, \rho] (2c_1 + c_2 + 2c_3 + 2(3c_1 + c_2 + c_3)r^{(1,0)}[\tau, \rho]^2)) + \\
& 4r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho])^2 (6(3c_1 + c_2 + c_3)r^{(0,1)}[\tau, \rho]^3\mathcal{A}^{(0,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) \\
& r^{(0,1)}[\tau, \rho] (\mathcal{A}^{(0,1)}[\tau, \rho] (10c_1 + 3c_2 + 2c_3 - 2(3c_1 + c_2 + c_3)r^{(1,0)}[\tau, \rho]^2) + \\
& 16(3c_1 + c_2 + c_3) (-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]r^{(1,1)}[\tau, \rho]) - \\
& 2(3c_1 + c_2 + c_3) (-1 + \mathcal{A}[\tau, \rho])r^{(0,1)}[\tau, \rho]^2 \\
& (6r^{(0,2)}[\tau, \rho] + r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - 2(-1 + \mathcal{A}[\tau, \rho])r^{(2,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho])^2 (2r^{(0,2)}[\tau, \rho] (-10c_1 - 3c_2 - 2c_3 + 2(3c_1 + c_2 + c_3)r^{(1,0)}[\tau, \rho]^2) + \\
& (10c_1 + 3c_2 + 2c_3)r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - 2(3c_1 + c_2 + c_3)r^{(1,0)}[\tau, \rho]^3 \\
& \mathcal{A}^{(1,0)}[\tau, \rho] + 2(10c_1 + 3c_2 + 2c_3) (-1 + \mathcal{A}[\tau, \rho])r^{(2,0)}[\tau, \rho] - \\
& 12(3c_1 + c_2 + c_3) (-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]^2r^{(2,0)}[\tau, \rho])) + \\
& r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho]) (-8c_1r^{(0,1)}[\tau, \rho]^2\mathcal{A}^{(0,1)}[\tau, \rho]^2 - 3c_2r^{(0,1)}[\tau, \rho]^2\mathcal{A}^{(0,1)}[\tau, \rho]^2 - \\
& 4c_3r^{(0,1)}[\tau, \rho]^2\mathcal{A}^{(0,1)}[\tau, \rho]^2 - 32c_1r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(0,2)}[\tau, \rho] - \\
& 12c_2r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(0,2)}[\tau, \rho] - 16c_3r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(0,2)}[\tau, \rho] + \\
& 32c_1\mathcal{A}[\tau, \rho]r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(0,2)}[\tau, \rho] + \\
& 12c_2\mathcal{A}[\tau, \rho]r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(0,2)}[\tau, \rho] + \\
& 16c_3\mathcal{A}[\tau, \rho]r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(0,2)}[\tau, \rho] - 32c_1r^{(0,2)}[\tau, \rho]^2 - \\
& 12c_2r^{(0,2)}[\tau, \rho]^2 - 16c_3r^{(0,2)}[\tau, \rho]^2 + 64c_1\mathcal{A}[\tau, \rho]r^{(0,2)}[\tau, \rho]^2 + \\
& 24c_2\mathcal{A}[\tau, \rho]r^{(0,2)}[\tau, \rho]^2 + 32c_3\mathcal{A}[\tau, \rho]r^{(0,2)}[\tau, \rho]^2 - 32c_1\mathcal{A}[\tau, \rho]^2r^{(0,2)}[\tau, \rho]^2 - \\
& 12c_2\mathcal{A}[\tau, \rho]^2r^{(0,2)}[\tau, \rho]^2 - 16c_3\mathcal{A}[\tau, \rho]^2r^{(0,2)}[\tau, \rho]^2 + 4\mathcal{G}_4[X_\delta] (-1 + \mathcal{A}[\tau, \rho])^3 \\
& (1 - r^{(0,1)}[\tau, \rho]^2 - r^{(1,0)}[\tau, \rho]^2 + \mathcal{A}[\tau, \rho] (-1 + r^{(1,0)}[\tau, \rho]^2)) + 8(-1 + \mathcal{A}[\tau, \rho])^2 \\
& \mathcal{G}_4'[X_\delta] (r^{(0,1)}[\tau, \rho]\phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho])^2 + \\
& 16c_1r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + 6c_2r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho] \\
& r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + 8c_3r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 16c_1\mathcal{A}[\tau, \rho]r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 6c_2\mathcal{A}[\tau, \rho]r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 8c_3\mathcal{A}[\tau, \rho]r^{(0,1)}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 32c_1r^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + 12c_2r^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 16c_3r^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - 64c_1\mathcal{A}[\tau, \rho]r^{(0,2)}[\tau, \rho] \\
& r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - 24c_2\mathcal{A}[\tau, \rho]r^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - \\
& 32c_3\mathcal{A}[\tau, \rho]r^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + 32c_1\mathcal{A}[\tau, \rho]^2r^{(0,2)}[\tau, \rho] \\
& r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + 12c_2\mathcal{A}[\tau, \rho]^2r^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + \\
& 16c_3\mathcal{A}[\tau, \rho]^2r^{(0,2)}[\tau, \rho]r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - 4c_1\mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 8c_1\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 4c_1\mathcal{A}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]^2 + 28c_1r^{(0,1)}[\tau, \rho]^2 \\
& \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2c_2r^{(0,1)}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 8c_3r^{(0,1)}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 28c_1\mathcal{A}[\tau, \rho]r^{(0,1)}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 28c_1\mathcal{A}[\tau, \rho]^2r^{(0,1)}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]^2 +
\end{aligned}$$

$$\begin{aligned}
& 8c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 4c_1 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& c_2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 4c_3 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 8c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 8c_3 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 4c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 4c_3 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 64c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - 32c_3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - \\
& 128c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 64c_3 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \\
& \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 64c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - \\
& 32c_3 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 64c_1 r^{(1,1)}[\tau, \rho]^2 - \\
& 32c_3 r^{(1,1)}[\tau, \rho]^2 - 192c_1 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho]^2 + 96c_3 \mathcal{A}[\tau, \rho] r^{(1,1)}[\tau, \rho]^2 + \\
& 192c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho]^2 - 96c_3 \mathcal{A}[\tau, \rho]^2 r^{(1,1)}[\tau, \rho]^2 - \\
& 64c_1 \mathcal{A}[\tau, \rho]^3 r^{(1,1)}[\tau, \rho]^2 + 32c_3 \mathcal{A}[\tau, \rho]^3 r^{(1,1)}[\tau, \rho]^2 - \\
& 64c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 12c_2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 128c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 24c_2 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho] \\
& \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 64c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 12c_2 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 128c_1 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 24c_2 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 384c_1 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 72c_2 \mathcal{A}[\tau, \rho] r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 384c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 72c_2 \mathcal{A}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 128c_1 \mathcal{A}[\tau, \rho]^3 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 24c_2 \mathcal{A}[\tau, \rho]^3 r^{(0,2)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 64c_1 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 12c_2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 192c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \\
& \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 36c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& 192c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 36c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \\
& \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 64c_1 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - \\
& 12c_2 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] - 32c_1 r^{(2,0)}[\tau, \rho]^2 - \\
& 12c_2 r^{(2,0)}[\tau, \rho]^2 - 16c_3 r^{(2,0)}[\tau, \rho]^2 + 128c_1 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho]^2 + \\
& 48c_2 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho]^2 + 64c_3 \mathcal{A}[\tau, \rho] r^{(2,0)}[\tau, \rho]^2 - 192c_1 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]^2 - \\
& 72c_2 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]^2 - 96c_3 \mathcal{A}[\tau, \rho]^2 r^{(2,0)}[\tau, \rho]^2 + \\
& 128c_1 \mathcal{A}[\tau, \rho]^3 r^{(2,0)}[\tau, \rho]^2 + 48c_2 \mathcal{A}[\tau, \rho]^3 r^{(2,0)}[\tau, \rho]^2 + \\
& 64c_3 \mathcal{A}[\tau, \rho]^3 r^{(2,0)}[\tau, \rho]^2 - 32c_1 \mathcal{A}[\tau, \rho]^4 r^{(2,0)}[\tau, \rho]^2 - 12c_2 \mathcal{A}[\tau, \rho]^4 r^{(2,0)}[\tau, \rho]^2 - \\
& 16c_3 \mathcal{A}[\tau, \rho]^4 r^{(2,0)}[\tau, \rho]^2 - 8c_1 \mathcal{A}^{(2,0)}[\tau, \rho] + 24c_1 \mathcal{A}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 24c_1 \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + 8c_1 \mathcal{A}[\tau, \rho]^3 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 24c_1 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + 4c_2 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 48c_1 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - 8c_2 \mathcal{A}[\tau, \rho] r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 24c_1 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + 4c_2 \mathcal{A}[\tau, \rho]^2 r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 24c_1 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + 4c_2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 72c_1 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - 12c_2 \mathcal{A}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 72c_1 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + 12c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 24c_1 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] - 4c_2 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& r[\tau, \rho]^3 (-8\mathcal{G}''[X_8] r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho]^4 \mathcal{A}^{(0,1)}[\tau, \rho] + \\
& 8\mathcal{A}[\tau, \rho] \mathcal{G}''[X_8] r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho]^4 \mathcal{A}^{(0,1)}[\tau, \rho] - 112c_1 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho]^3 - \\
& 42c_2 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho]^3 - 56c_3 r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho]^3 - \\
& 152c_1 \mathcal{A}^{(0,1)}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho] - 57c_2 \mathcal{A}^{(0,1)}[\tau, \rho]^2 r^{(0,2)}[\tau, \rho] - \\
& 76c_3 \mathcal{A}^{(0,1)}[\tau, \rho]^2 r^{(0,2)}[\$$

[illegible]

[illegible]

$$\begin{aligned}
& 36 c_2 \mathcal{A}[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 96 c_1 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - \\
& 24 c_2 \mathcal{A}[\tau, \rho]^3 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + 24 c_1 \mathcal{A}[\tau, \rho]^4 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] + \\
& 6 c_2 \mathcal{A}[\tau, \rho]^4 r^{(1,0)}[\tau, \rho] \mathcal{A}^{(3,0)}[\tau, \rho] - 32 c_1 r^{(4,0)}[\tau, \rho] - 12 c_2 r^{(4,0)}[\tau, \rho] - \\
& 16 c_3 r^{(4,0)}[\tau, \rho] + 160 c_1 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] + 60 c_2 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] + \\
& 80 c_3 \mathcal{A}[\tau, \rho] r^{(4,0)}[\tau, \rho] - 320 c_1 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] - 120 c_2 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] - \\
& 160 c_3 \mathcal{A}[\tau, \rho]^2 r^{(4,0)}[\tau, \rho] + 320 c_1 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] + \\
& 120 c_2 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] + 160 c_3 \mathcal{A}[\tau, \rho]^3 r^{(4,0)}[\tau, \rho] - \\
& 160 c_1 \mathcal{A}[\tau, \rho]^4 r^{(4,0)}[\tau, \rho] - 60 c_2 \mathcal{A}[\tau, \rho]^4 r^{(4,0)}[\tau, \rho] - 80 c_3 \mathcal{A}[\tau, \rho]^4 r^{(4,0)}[\tau, \rho] + \\
& 32 c_1 \mathcal{A}[\tau, \rho]^5 r^{(4,0)}[\tau, \rho] + 12 c_2 \mathcal{A}[\tau, \rho]^5 r^{(4,0)}[\tau, \rho] + 16 c_3 \mathcal{A}[\tau, \rho]^5 r^{(4,0)}[\tau, \rho] + \\
& r[\tau, \rho]^4 (-4 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{G}_4''[X_\delta] (\phi^{(0,1)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho]^2) \\
& (\phi^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(0,1)}[\tau, \rho] \\
& (2 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) + 2 (-1 + \mathcal{A}[\tau, \rho])^2 \\
& (2 \phi^{(1,1)}[\tau, \rho]^2 + (-2 \phi^{(0,2)}[\tau, \rho] + \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) \phi^{(2,0)}[\tau, \rho])) + \\
& 2 (-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{G}_4'[X_\delta] (-4 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(0,1)}[\tau, \rho] (2 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - \\
& \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) + \phi^{(0,1)}[\tau, \rho]^2 (3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2 (-1 + \mathcal{A}[\tau, \rho]) \\
& \mathcal{A}^{(2,0)}[\tau, \rho]) + (-1 + \mathcal{A}[\tau, \rho]) (4 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \\
& \phi^{(2,0)}[\tau, \rho] + 8 (-1 + \mathcal{A}[\tau, \rho]) (\phi^{(1,1)}[\tau, \rho]^2 - \phi^{(0,2)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) - \\
& \phi^{(1,0)}[\tau, \rho]^2 (\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]))) + \\
& (4 c_1 + c_2) (\mathcal{A}^{(0,1)}[\tau, \rho]^2 (7 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 5 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(0,1)}[\tau, \rho] (-9 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + \\
& 5 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,1)}[\tau, \rho]) - (-1 + \mathcal{A}[\tau, \rho]) (5 \mathcal{A}^{(1,0)}[\tau, \rho]^4 - \\
& 12 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + 2 \mathcal{A}^{(0,2)}[\tau, \rho] \\
& (\mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) + (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho] \\
& (-2 \mathcal{A}^{(1,2)}[\tau, \rho] + 5 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(3,0)}[\tau, \rho]) - 2 (-1 + \mathcal{A}[\tau, \rho]) \\
& (\mathcal{A}^{(1,1)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) (2 \mathcal{A}^{(2,0)}[\tau, \rho]^2 + \mathcal{A}^{(2,2)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \\
& \mathcal{A}^{(4,0)}[\tau, \rho]))) / (4 r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho])^5) == 0
\end{aligned}$$

Scalar field equation components

Euclidean scalar field equation (component form):

$$\begin{aligned}
& \frac{4 \mathcal{K}[X_\delta] \left(-\frac{r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho]}{-1 + \mathcal{A}[\tau, \rho]} + r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \right)}{r[\tau, \rho]} - \\
& \frac{\mathcal{K}[X_\delta] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) (2 \phi^{(0,2)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]))}{(-1 + \mathcal{A}[\tau, \rho])^2} + \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^3} \\
& 2 \mathcal{K}''[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) (2 \phi^{(0,2)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho])) + \\
& \frac{1}{r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho])^4} 12 \mathcal{G}_4''[X_\delta] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho])^2 \\
& (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) (2 \phi^{(0,2)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho])) - \frac{1}{r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho])^5} \\
& 8 \mathcal{G}_4^{(3)}[X_\delta] \phi^{(0,1)}[\tau, \rho]^2 (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho])^2 \\
& (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) (2 \phi^{(0,2)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho])) - \\
& \frac{4 \mathcal{K}''[X_\delta] \phi^{(0,1)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho])}{(-1 + \mathcal{A}[\tau, \rho])^2} + \frac{1}{r[\tau, \rho]^2 (-1 + \mathcal{A}[\tau, \rho])^4} \\
& 16 \mathcal{G}_4^{(3)}[X_\delta] \phi^{(0,1)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho])^2 \\
& (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) - \frac{1}{r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho])^4} \\
& 12 \mathcal{G}_4''[X_\delta] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho])
\end{aligned}$$

[illegible]

$$\begin{aligned}
& 2(-1+\mathcal{A}[\tau, \rho])^2 \left((-1+\mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] (-r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) \right) + \\
& \frac{1}{r[\tau, \rho]^2} 16 \mathcal{G}_4'[X_\delta] \phi^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] \left(\frac{1}{2(-1+\mathcal{A}[\tau, \rho])^2} r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \right. \\
& \quad \left. (2(-1+\mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] + r[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - 2r[\tau, \rho] (-1+\mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho] + \right. \\
& \quad \left. r^{(1,0)}[\tau, \rho] (-r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) \right) + \frac{1}{r[\tau, \rho]^3} \\
& 8 \mathcal{G}_4'[X_\delta] \left(-\frac{r^{(0,1)}[\tau, \rho]^3 \phi^{(0,1)}[\tau, \rho]}{(-1+\mathcal{A}[\tau, \rho])^2} + \frac{1}{2(-1+\mathcal{A}[\tau, \rho])^3} r^{(0,1)}[\tau, \rho]^2 \right. \\
& \quad \left(-r[\tau, \rho] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] + 2(-1+\mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] (-1+\mathcal{A}[\tau, \rho]) \right. \\
& \quad \left. (2\phi^{(0,2)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) \right) + \frac{1}{(-1+\mathcal{A}[\tau, \rho])^2} r^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho] \\
& \quad \left(\phi^{(0,1)}[\tau, \rho] ((-1+\mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] + r[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - 2r[\tau, \rho] (-1+\mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho] + \right. \\
& \quad \left. r^{(1,0)}[\tau, \rho]^2 (-r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) \right) - \frac{1}{r[\tau, \rho]^3} \\
& 8 \mathcal{G}_4'[X_\delta] \left(-\frac{r^{(0,1)}[\tau, \rho]^3 \phi^{(0,1)}[\tau, \rho]}{(-1+\mathcal{A}[\tau, \rho])^2} + \frac{1}{(-1+\mathcal{A}[\tau, \rho])^3} r^{(0,1)}[\tau, \rho]^2 (-r[\tau, \rho] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] + r[\tau, \rho] \right. \\
& \quad \left. (-1+\mathcal{A}[\tau, \rho]) \phi^{(0,2)}[\tau, \rho] + (-1+\mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] \right) + \frac{1}{(-1+\mathcal{A}[\tau, \rho])^2} r^{(0,1)}[\tau, \rho] \\
& \quad \left((-1+\mathcal{A}[\tau, \rho]) \phi^{(0,1)}[\tau, \rho] r^{(1,0)}[\tau, \rho]^2 + r[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (r^{(0,2)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - \right. \\
& \quad \left. (-1+\mathcal{A}[\tau, \rho]) (\phi^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 2r^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho])) \right) + \\
& \quad r^{(1,0)}[\tau, \rho] (-r^{(1,0)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] (-\frac{\phi^{(0,1)}[\tau, \rho] r^{(1,1)}[\tau, \rho]}{-1+\mathcal{A}[\tau, \rho]} + \phi^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho]) + \\
& \quad \left. r[\tau, \rho] r^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) \right) - \frac{1}{r[\tau, \rho]^3 (-1+\mathcal{A}[\tau, \rho])^4} 2 \mathcal{G}_4'[X_\delta] \\
& (-2r[\tau, \rho] (-1+\mathcal{A}[\tau, \rho])^2 r^{(0,2)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) + \\
& 4(1-\mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho] \\
& \quad ((1-\mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) - \\
& \quad r[\tau, \rho] (r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) \phi^{(0,2)}[\tau, \rho]) + \\
& \quad (-1+\mathcal{A}[\tau, \rho]) (-\phi^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + (-1+\mathcal{A}[\tau, \rho]) \\
& \quad (\phi^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho]))) - r[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \\
& ((1-\mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) - \\
& \quad r[\tau, \rho] (r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) \phi^{(0,2)}[\tau, \rho]) + \\
& \quad (-1+\mathcal{A}[\tau, \rho]) (-\phi^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + \\
& \quad (-1+\mathcal{A}[\tau, \rho]) (\phi^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho]))) + \\
& 2r[\tau, \rho]^2 (2r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho]^2 - (-1+\mathcal{A}[\tau, \rho]) \mathcal{A}^{(0,1)}[\tau, \rho] \\
& \quad (\phi^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + r^{(0,1)}[\tau, \rho] \phi^{(0,2)}[\tau, \rho]) + (1-\mathcal{A}[\tau, \rho]) (r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \\
& \quad \phi^{(0,2)}[\tau, \rho] + \phi^{(0,1)}[\tau, \rho] (\mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,2)}[\tau, \rho])) + \\
& \quad (-1+\mathcal{A}[\tau, \rho])^2 (2r^{(0,2)}[\tau, \rho] \phi^{(0,2)}[\tau, \rho] + \phi^{(0,1)}[\tau, \rho] r^{(0,3)}[\tau, \rho] + r^{(0,1)}[\tau, \rho] \phi^{(0,3)}[\tau, \rho]) - \\
& \quad 2(-1+\mathcal{A}[\tau, \rho])^3 r^{(1,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho])^3 \phi^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] - \\
& \quad (-1+\mathcal{A}[\tau, \rho])^3 r^{(1,0)}[\tau, \rho] \phi^{(1,2)}[\tau, \rho] + \\
& r[\tau, \rho] (1-\mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho] ((-1+\mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho] - r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \\
& \quad ((-1+\mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] + r[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - r[\tau, \rho] (-1+\mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) - \\
& \quad r[\tau, \rho] (1-\mathcal{A}[\tau, \rho]) (\phi^{(0,1)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho]) - \\
& \quad r[\tau, \rho] (-1+\mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) - \\
& \frac{1}{r[\tau, \rho]^3 (-1+\mathcal{A}[\tau, \rho])^4} 2 \mathcal{G}_4'[X_\delta] (4(1-\mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho] \\
& \quad ((1-\mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) - \\
& \quad r[\tau, \rho] (r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1+\mathcal{A}[\tau, \rho]) \phi^{(0,2)}[\tau, \rho]) + \\
& \quad (-1+\mathcal{A}[\tau, \rho]) (-\phi^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + (-1+\mathcal{A}[\tau, \rho]) \\
& \quad (\phi^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho]))) - r[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho]
\end{aligned}$$

$$\begin{aligned}
& ((1 - \mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) - \\
& r[\tau, \rho] (r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(0,2)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho]) (-\phi^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + \\
& (-1 + \mathcal{A}[\tau, \rho]) (\phi^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho])) - 2 r[\tau, \rho] \\
& ((-1 + \mathcal{A}[\tau, \rho])^2 r^{(0,2)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) - r[\tau, \rho] \\
& (2 r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + \\
& r^{(0,1)}[\tau, \rho] \phi^{(0,2)}[\tau, \rho]) + (1 - \mathcal{A}[\tau, \rho]) (r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(0,2)}[\tau, \rho] + \\
& \phi^{(0,1)}[\tau, \rho] (\mathcal{A}^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,2)}[\tau, \rho])) + (-1 + \mathcal{A}[\tau, \rho])^2 \\
& (2 r^{(0,2)}[\tau, \rho] \phi^{(0,2)}[\tau, \rho] + \phi^{(0,1)}[\tau, \rho] r^{(0,3)}[\tau, \rho] + r^{(0,1)}[\tau, \rho] \phi^{(0,3)}[\tau, \rho]) - \\
& 2 (-1 + \mathcal{A}[\tau, \rho])^3 r^{(1,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])^3 \phi^{(1,0)}[\tau, \rho] r^{(1,2)}[\tau, \rho] - \\
& (-1 + \mathcal{A}[\tau, \rho])^3 r^{(1,0)}[\tau, \rho] \phi^{(1,2)}[\tau, \rho])) + \\
& r[\tau, \rho] (1 - \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho] ((-1 + \mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho]^2 \phi^{(1,0)}[\tau, \rho] - r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \\
& ((-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] + r[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) - \\
& r[\tau, \rho] (1 - \mathcal{A}[\tau, \rho]) (\phi^{(0,1)}[\tau, \rho] r^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho]) - \\
& r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho])) + \frac{1}{r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho])^4} \\
& 4 \mathcal{G}_4''[X_\varepsilon] \phi^{(1,0)}[\tau, \rho] (-r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) \\
& (2 \phi^{(0,1)}[\tau, \rho] (2 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,1)}[\tau, \rho]) - (-1 + \mathcal{A}[\tau, \rho]) \\
& (4 \phi^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2 \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] + 4 \phi^{(1,2)}[\tau, \rho] - \\
& 4 \mathcal{A}[\tau, \rho] \phi^{(1,2)}[\tau, \rho] - 2 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + 2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho])) - \\
& \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^5} \mathcal{G}_4''[X_\varepsilon] \phi^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) \\
& (2 \phi^{(0,1)}[\tau, \rho] (2 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,1)}[\tau, \rho]) - (-1 + \mathcal{A}[\tau, \rho]) \\
& (4 \phi^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2 \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] + 4 \phi^{(1,2)}[\tau, \rho] - \\
& 4 \mathcal{A}[\tau, \rho] \phi^{(1,2)}[\tau, \rho] - 2 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + 2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho])) - \\
& \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^3} 2 \mathcal{G}_4''[X_\varepsilon] \phi^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] \\
& (2 \phi^{(0,1)}[\tau, \rho] (2 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,1)}[\tau, \rho]) - \\
& (-1 + \mathcal{A}[\tau, \rho]) (4 \phi^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 2 \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] + 4 \phi^{(1,2)}[\tau, \rho] - 4 \mathcal{A}[\tau, \rho] \phi^{(1,2)}[\tau, \rho] - \\
& 2 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + 2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho])) + \frac{1}{r[\tau, \rho]^3 (1 - \mathcal{A}[\tau, \rho])} \\
& 2 \mathcal{G}_4'[X_\varepsilon] (-4 r^{(0,1)}[\tau, \rho] (\frac{r^{(0,1)}[\tau, \rho]^2 \phi^{(0,1)}[\tau, \rho]}{-1 + \mathcal{A}[\tau, \rho]} + \frac{1}{2} r^{(0,1)}[\tau, \rho] (-2 r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^2} r[\tau, \rho] \\
& (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) (2 \phi^{(0,2)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]))) + \\
& r[\tau, \rho] r^{(1,0)}[\tau, \rho] (\frac{\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]}{2 - 2 \mathcal{A}[\tau, \rho]} + \phi^{(1,1)}[\tau, \rho])) - \\
& 4 r^{(0,1)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] (\frac{r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho]}{-1 + \mathcal{A}[\tau, \rho]} - r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho]) + \\
& r[\tau, \rho] (\frac{r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho]}{(-1 + \mathcal{A}[\tau, \rho])^2} + \frac{\phi^{(0,1)}[\tau, \rho] r^{(0,2)}[\tau, \rho] + r^{(0,1)}[\tau, \rho] \phi^{(0,2)}[\tau, \rho]}{1 - \mathcal{A}[\tau, \rho]} + \\
& \phi^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho])) - \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^3} \\
& 2 r[\tau, \rho]^2 r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (2 \mathcal{A}^{(0,1)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) (\mathcal{A}^{(0,2)}[\tau, \rho] - \mathcal{A}^{(1,0)}[\tau, \rho]^2)) + \\
& (-1 + \mathcal{A}[\tau, \rho]) (\mathcal{A}^{(0,1)}[\tau, \rho] (-3 \phi^{(0,2)}[\tau, \rho] + \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho]) (2 \phi^{(0,3)}[\tau, \rho] - 3 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho]))) + \\
& \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^2} r[\tau, \rho]^2 r^{(1,0)}[\tau, \rho] (2 \phi^{(0,1)}[\tau, \rho] (2 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,1)}[\tau, \rho]) - \\
& (-1 + \mathcal{A}[\tau, \rho]) (4 \phi^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 + \\
& 2 \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] + 4 \phi^{(1,2)}[\tau, \rho] - 4 \mathcal{A}[\tau, \rho] \phi^{(1,2)}[\tau, \rho] -
\end{aligned}$$

$$\begin{aligned}
& 4 \mathcal{G}_4'[\mathcal{X}_\varepsilon] \psi^{(0,1)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] \psi^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \psi^{(0,1)}[\tau, \rho]) \\
& (\phi^{(0,1)}[\tau, \rho] (-3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& 4 (-1 + \mathcal{A}[\tau, \rho]) (\mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(2,1)}[\tau, \rho])) + \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^5} \\
& \mathcal{G}_4''[\mathcal{X}_\varepsilon] \phi^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) (2 \phi^{(0,2)}[\tau, \rho] - \phi^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho])) \\
& (\phi^{(0,1)}[\tau, \rho] (-3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& 4 (-1 + \mathcal{A}[\tau, \rho]) (\mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(2,1)}[\tau, \rho])) - \\
& \frac{1}{r[\tau, \rho]^3 (-1 + \mathcal{A}[\tau, \rho])^4} 2 \mathcal{G}_4'[\mathcal{X}_\varepsilon] \phi^{(0,1)}[\tau, \rho] (4 (-1 + \mathcal{A}[\tau, \rho])^2 r^{(0,1)}[\tau, \rho]^3 - \\
& 2 r^{(0,1)}[\tau, \rho] (2 + r[\tau, \rho]^2 (2 \mathcal{A}^{(0,1)}[\tau, \rho]^2 + \mathcal{A}^{(0,2)}[\tau, \rho]) - 2 r^{(1,0)}[\tau, \rho]^2 - \mathcal{A}[\tau, \rho] \\
& (6 + r[\tau, \rho]^2 \mathcal{A}^{(0,2)}[\tau, \rho] - 6 r^{(1,0)}[\tau, \rho]^2 + 2 r[\tau, \rho] (r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 3 r^{(2,0)}[\tau, \rho])) + \\
& r[\tau, \rho] (r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2 r^{(2,0)}[\tau, \rho]) + \\
& \mathcal{A}[\tau, \rho]^2 (6 - 6 r^{(1,0)}[\tau, \rho]^2 + r[\tau, \rho] r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 6 r[\tau, \rho] r^{(2,0)}[\tau, \rho]) + \\
& 2 \mathcal{A}[\tau, \rho]^3 (-1 + r^{(1,0)}[\tau, \rho]^2 + r[\tau, \rho] r^{(2,0)}[\tau, \rho])) + \\
& r[\tau, \rho] (4 (-1 + \mathcal{A}[\tau, \rho])^3 r^{(1,0)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + 2 r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho]) (\mathcal{A}^{(0,1)}[\tau, \rho] \\
& (3 r^{(0,2)}[\tau, \rho] - r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) + (-1 + \mathcal{A}[\tau, \rho]) (-2 r^{(0,3)}[\tau, \rho] + \mathcal{A}^{(1,0)}[\tau, \rho] \\
& r^{(1,1)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - 2 r^{(2,1)}[\tau, \rho] + 2 \mathcal{A}[\tau, \rho] r^{(2,1)}[\tau, \rho])) + \\
& r[\tau, \rho]^2 (\mathcal{A}^{(0,1)}[\tau, \rho] (\mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) - \\
& (-1 + \mathcal{A}[\tau, \rho]) (\mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,1)}[\tau, \rho])))) - \\
& \frac{1}{r[\tau, \rho]^2} 4 \mathcal{G}_4'[\mathcal{X}_\varepsilon] (-\frac{1}{r[\tau, \rho]} 2 r^{(1,0)}[\tau, \rho] (\frac{1}{2 (-1 + \mathcal{A}[\tau, \rho])^2} r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (2 (-1 + \mathcal{A}[\tau, \rho]) \\
& r^{(1,0)}[\tau, \rho] + r[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - 2 r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) + \\
& r^{(1,0)}[\tau, \rho] (-r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] \phi^{(2,0)}[\tau, \rho])) - \frac{1}{-1 + \mathcal{A}[\tau, \rho]} \\
& r[\tau, \rho] r^{(0,1)}[\tau, \rho] (\frac{\mathcal{A}^{(1,0)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 2 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho])}{2 (-1 + \mathcal{A}[\tau, \rho])^2} + \phi^{(2,1)}[\tau, \rho]) + \\
& r[\tau, \rho] r^{(1,0)}[\tau, \rho] \phi^{(3,0)}[\tau, \rho]) + \frac{1}{r[\tau, \rho]^2} \\
& 4 \mathcal{G}_4'[\mathcal{X}_\varepsilon] (\frac{2 r^{(1,0)}[\tau, \rho]^2 (-r^{(0,1)}[\tau, \rho] \phi^{(0,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho])}{r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho])} - \frac{1}{2 (-1 + \mathcal{A}[\tau, \rho])^2} \\
& r^{(1,0)}[\tau, \rho] (r^{(0,1)}[\tau, \rho] (3 \phi^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - 4 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) + \\
& 2 (-1 + \mathcal{A}[\tau, \rho]) (-\phi^{(0,1)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + \\
& (-1 + \mathcal{A}[\tau, \rho]) (\phi^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + 2 r^{(1,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho])) - \frac{1}{4 (-1 + \mathcal{A}[\tau, \rho])^3} \\
& r[\tau, \rho] r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) - 4 (-1 + \mathcal{A}[\tau, \rho]) \\
& (\mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \phi^{(2,1)}[\tau, \rho])) + r[\tau, \rho] r^{(1,0)}[\tau, \rho] \phi^{(3,0)}[\tau, \rho]) + \\
& \frac{1}{r[\tau, \rho]^2} 8 \mathcal{G}_4'[\mathcal{X}_\varepsilon] (r^{(2,0)}[\tau, \rho] (-r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] \phi^{(2,0)}[\tau, \rho]) - \frac{1}{r[\tau, \rho]} \\
& 2 r^{(1,0)}[\tau, \rho] (\frac{1}{2 (-1 + \mathcal{A}[\tau, \rho])^2} r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (2 (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] + r[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - \\
& 2 r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) + \\
& r^{(1,0)}[\tau, \rho] (-r^{(1,0)}[\tau, \rho] \phi^{(1,0)}[\tau, \rho] + r[\tau, \rho] \phi^{(2,0)}[\tau, \rho])) + \frac{1}{2 (-1 + \mathcal{A}[\tau, \rho])^3} \\
& ((-1 + \mathcal{A}[\tau, \rho]) r^{(1,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (2 (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] + r[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) - \\
& 2 r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho]) \phi^{(1,1)}[\tau, \rho]) + \\
& r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (-(-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]) + 2 (-1 + \mathcal{A}[\tau, \rho])^2 r^{(2,0)}[\tau, \rho] + \\
& r[\tau, \rho] (-2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 + (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho])) + \\
& r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho]) (3 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - 2 (-1 + \mathcal{A}[\tau, \rho]) \phi^{(2,1)}[\tau, \rho])))) + \\
& r^{(1,0)}[\tau, \rho] (-\phi^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + r[\tau, \rho] \phi^{(3,0)}[\tau, \rho])) - \frac{1}{r[\tau, \rho]^3} \\
& 4 \mathcal{G}_4'[\mathcal{X}_\varepsilon] (-\frac{1}{2 (-1 + \mathcal{A}[\tau, \rho])^3} r^{(0,1)}[\tau, \rho] (\phi^{(0,1)}[\tau, \rho] (4 (-1 + \mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho]^2 +
\end{aligned}$$

$$\begin{aligned}
& 2r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + r[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]^2 - \\
& 2r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])(2(-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] + \\
& r[\tau, \rho](\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])\phi^{(2,1)}[\tau, \rho])) + \\
& r^{(1,0)}[\tau, \rho](2r^{(1,0)}[\tau, \rho]^2\phi^{(1,0)}[\tau, \rho] - 2r[\tau, \rho]r^{(1,0)}[\tau, \rho]\phi^{(2,0)}[\tau, \rho] + r[\tau, \rho]^2\phi^{(3,0)}[\tau, \rho])) + \\
& \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^4}\mathcal{G}_4'[X_\varepsilon](2\phi^{(0,1)}[\tau, \rho](2\mathcal{A}^{(0,1)}[\tau, \rho](3\mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho])(-4\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,1)}[\tau, \rho])) + \\
& (-1 + \mathcal{A}[\tau, \rho])(-8\mathcal{A}^{(0,1)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] - 4\phi^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] + 4\mathcal{A}[\tau, \rho] \\
& \phi^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] - 8\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,2)}[\tau, \rho] + 8\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,2)}[\tau, \rho] + \\
& 3\mathcal{A}^{(1,0)}[\tau, \rho]^2\phi^{(2,0)}[\tau, \rho] - 3\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^2\phi^{(2,0)}[\tau, \rho] + 2\phi^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] - \\
& 4\mathcal{A}[\tau, \rho]\phi^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] + 2\mathcal{A}[\tau, \rho]^2\phi^{(2,0)}[\tau, \rho]\mathcal{A}^{(2,0)}[\tau, \rho] + \\
& 2\phi^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho](\mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) - \\
& 4\phi^{(0,2)}[\tau, \rho](2\mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) - \\
& 2\mathcal{A}^{(0,1)}[\tau, \rho]\phi^{(2,1)}[\tau, \rho] + 2\mathcal{A}[\tau, \rho]\mathcal{A}^{(0,1)}[\tau, \rho]\phi^{(2,1)}[\tau, \rho] - 4\phi^{(2,2)}[\tau, \rho] + \\
& 8\mathcal{A}[\tau, \rho]\phi^{(2,2)}[\tau, \rho] - 4\mathcal{A}[\tau, \rho]^2\phi^{(2,2)}[\tau, \rho] + 2\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(3,0)}[\tau, \rho] - \\
& 4\mathcal{A}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(3,0)}[\tau, \rho] + 2\mathcal{A}[\tau, \rho]^2\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(3,0)}[\tau, \rho])) + \frac{1}{r[\tau, \rho]^3} \\
& \mathcal{G}_4'[X_\varepsilon](-\frac{1}{(-1 + \mathcal{A}[\tau, \rho])^3}r^{(0,1)}[\tau, \rho](\phi^{(0,1)}[\tau, \rho](8(-1 + \mathcal{A}[\tau, \rho])^2r^{(1,0)}[\tau, \rho]^2 + 6r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho]) \\
& r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] + r[\tau, \rho]^2(3\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho])) - \\
& 4r[\tau, \rho](-1 + \mathcal{A}[\tau, \rho])(2(-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] + \\
& r[\tau, \rho](\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])\phi^{(2,1)}[\tau, \rho])) + \\
& 4r^{(1,0)}[\tau, \rho](2r^{(1,0)}[\tau, \rho]^2\phi^{(1,0)}[\tau, \rho] - 2r[\tau, \rho]r^{(1,0)}[\tau, \rho]\phi^{(2,0)}[\tau, \rho] + \\
& r[\tau, \rho](\frac{\phi^{(0,1)}[\tau, \rho]r^{(1,1)}[\tau, \rho]}{-1 + \mathcal{A}[\tau, \rho]} - \phi^{(1,0)}[\tau, \rho]r^{(2,0)}[\tau, \rho] + r[\tau, \rho]\phi^{(3,0)}[\tau, \rho])) + \\
& (\mathcal{G}_4'[X_\varepsilon](-\phi^{(0,1)}[\tau, \rho](\mathcal{A}^{(0,1)}[\tau, \rho](23\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 6(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& 4(-1 + \mathcal{A}[\tau, \rho])(-4\mathcal{A}^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,1)}[\tau, \rho])) + \\
& (-1 + \mathcal{A}[\tau, \rho])(2\phi^{(0,2)}[\tau, \rho](7\mathcal{A}^{(1,0)}[\tau, \rho]^2 - 2(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& \phi^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho](-3\mathcal{A}^{(1,0)}[\tau, \rho]^2 + 2(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& 4\mathcal{A}^{(0,1)}[\tau, \rho](4\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])\phi^{(2,1)}[\tau, \rho]) - \\
& 4(-1 + \mathcal{A}[\tau, \rho])(2\phi^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] - \mathcal{A}^{(1,0)}[\tau, \rho]^2\phi^{(2,0)}[\tau, \rho] - 2(-1 + \mathcal{A}[\tau, \rho]) \\
& \phi^{(2,2)}[\tau, \rho] + \mathcal{A}^{(1,0)}[\tau, \rho](4\phi^{(1,2)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])\phi^{(3,0)}[\tau, \rho])))) / \\
& (2(1 - \mathcal{A}[\tau, \rho])(-1 + \mathcal{A}[\tau, \rho])^3) - \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^4}\mathcal{G}_4'[X_\varepsilon](3\phi^{(0,1)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] \\
& (3\mathcal{A}^{(0,1)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - 2(-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(1,1)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho])(-6\phi^{(0,2)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^2 + \phi^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho]^3 - \\
& 2\mathcal{A}^{(0,1)}[\tau, \rho](4\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])\phi^{(2,1)}[\tau, \rho]) + 2(-1 + \mathcal{A}[\tau, \rho]) \\
& (2\phi^{(1,1)}[\tau, \rho]\mathcal{A}^{(1,1)}[\tau, \rho] - \mathcal{A}^{(1,0)}[\tau, \rho]^2\phi^{(2,0)}[\tau, \rho] - 2(-1 + \mathcal{A}[\tau, \rho])\phi^{(2,2)}[\tau, \rho] + \\
& \mathcal{A}^{(1,0)}[\tau, \rho](4\phi^{(1,2)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])\phi^{(3,0)}[\tau, \rho])))) - \frac{1}{r[\tau, \rho]^2(-1 + \mathcal{A}[\tau, \rho])^3}8\mathcal{G}_4'[X_\varepsilon] \\
& (\frac{1}{r[\tau, \rho]}2(-1 + \mathcal{A}[\tau, \rho])^2r^{(1,0)}[\tau, \rho]^2(-r^{(0,1)}[\tau, \rho]\phi^{(0,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]\phi^{(1,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho])(r^{(0,1)}[\tau, \rho](2(-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] - \\
& \phi^{(0,1)}[\tau, \rho](2r^{(1,0)}[\tau, \rho]\mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])r^{(2,0)}[\tau, \rho])) - \\
& (-1 + \mathcal{A}[\tau, \rho])r^{(1,0)}[\tau, \rho](-2\phi^{(0,1)}[\tau, \rho]r^{(1,1)}[\tau, \rho] + \\
& (-1 + \mathcal{A}[\tau, \rho])(3\phi^{(1,0)}[\tau, \rho]r^{(2,0)}[\tau, \rho] + 2r^{(1,0)}[\tau, \rho]\phi^{(2,0)}[\tau, \rho])) - \\
& r[\tau, \rho](r^{(0,1)}[\tau, \rho](\phi^{(0,1)}[\tau, \rho](2\mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho])\mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& (-1 + \mathcal{A}[\tau, \rho])(-2\mathcal{A}^{(1,0)}[\tau, \rho]\phi^{(1,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])\phi^{(2,1)}[\tau, \rho])) - \\
& (-1 + \mathcal{A}[\tau, \rho])(\phi^{(0,1)}[\tau, \rho](2\mathcal{A}^{(1,0)}[\tau, \rho]r^{(1,1)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho])r^{(2,1)}[\tau, \rho]) +
\end{aligned}$$

$$\begin{aligned}
& (-1 + \mathcal{A}[\tau, \rho]) (-2 r^{(1,1)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) (2 r^{(2,0)}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] + \\
& \quad \phi^{(1,0)}[\tau, \rho] r^{(3,0)}[\tau, \rho] + r^{(1,0)}[\tau, \rho] \phi^{(3,0)}[\tau, \rho])) - \\
& \frac{1}{(-1 + \mathcal{A}[\tau, \rho])^4} 2 \mathcal{G}_4'[X_\mathcal{E}] (\phi^{(0,1)}[\tau, \rho] (2 \mathcal{A}^{(0,1)}[\tau, \rho] (3 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) + \\
& \quad (-1 + \mathcal{A}[\tau, \rho]) (-4 \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,1)}[\tau, \rho])) + \\
& \quad (-1 + \mathcal{A}[\tau, \rho]) (-4 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] - 2 \phi^{(1,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] + \\
& \quad 2 \mathcal{A}[\tau, \rho] \phi^{(1,1)}[\tau, \rho] \mathcal{A}^{(1,1)}[\tau, \rho] - 4 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,2)}[\tau, \rho] + \\
& \quad 4 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(1,2)}[\tau, \rho] + 2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 \phi^{(2,0)}[\tau, \rho] - 2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho]^2 \\
& \quad \phi^{(2,0)}[\tau, \rho] + 2 \phi^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 4 \mathcal{A}[\tau, \rho] \phi^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& \quad 2 \mathcal{A}[\tau, \rho]^2 \phi^{(2,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] - 2 \phi^{(0,2)}[\tau, \rho] (2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(2,0)}[\tau, \rho]) - \\
& \quad \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,1)}[\tau, \rho] + \mathcal{A}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] \phi^{(2,1)}[\tau, \rho] - 2 \phi^{(2,2)}[\tau, \rho] + \\
& \quad 4 \mathcal{A}[\tau, \rho] \phi^{(2,2)}[\tau, \rho] - 2 \mathcal{A}[\tau, \rho]^2 \phi^{(2,2)}[\tau, \rho] + \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(3,0)}[\tau, \rho] - \\
& \quad 2 \mathcal{A}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(3,0)}[\tau, \rho] + \mathcal{A}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] \phi^{(3,0)}[\tau, \rho] + \phi^{(1,0)}[\tau, \rho] \\
& \quad (2 \mathcal{A}^{(1,0)}[\tau, \rho]^3 - 3 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{A}^{(3,0)}[\tau, \rho])) + \\
& \frac{1}{r[\tau, \rho]^3 (-1 + \mathcal{A}[\tau, \rho])^3} 2 \mathcal{G}_4'[X_\mathcal{E}] \phi^{(1,0)}[\tau, \rho] (-4 (-1 + \mathcal{A}[\tau, \rho])^3 r^{(1,0)}[\tau, \rho]^3 - \\
& \quad 2 r[\tau, \rho] (-1 + \mathcal{A}[\tau, \rho])^2 r^{(1,0)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] + \\
& \quad 2 (-1 + \mathcal{A}[\tau, \rho]) r^{(1,0)}[\tau, \rho] (2 + 2 \mathcal{A}[\tau, \rho]^2 - 2 r^{(0,1)}[\tau, \rho]^2 - r[\tau, \rho] r^{(0,1)}[\tau, \rho] \mathcal{A}^{(0,1)}[\tau, \rho] - \\
& \quad 2 r[\tau, \rho] r^{(0,2)}[\tau, \rho] - r[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho]^2 - r[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho] + \\
& \quad \mathcal{A}[\tau, \rho] (-4 + 2 r^{(0,1)}[\tau, \rho]^2 + 2 r[\tau, \rho] r^{(0,2)}[\tau, \rho] + r[\tau, \rho]^2 \mathcal{A}^{(2,0)}[\tau, \rho])) + \\
& \quad r[\tau, \rho] (2 (-1 + \mathcal{A}[\tau, \rho]) r^{(0,1)}[\tau, \rho]^2 \mathcal{A}^{(1,0)}[\tau, \rho] - 2 r^{(0,1)}[\tau, \rho] (2 (-1 + \mathcal{A}[\tau, \rho])^2 r^{(1,1)}[\tau, \rho] + \\
& \quad r[\tau, \rho] (2 \mathcal{A}^{(0,1)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] - (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,1)}[\tau, \rho])) + \\
& \quad r[\tau, \rho] (4 (-1 + \mathcal{A}[\tau, \rho]) r^{(0,2)}[\tau, \rho] \mathcal{A}^{(1,0)}[\tau, \rho] + 2 (-1 + \mathcal{A}[\tau, \rho]) \\
& \quad (\mathcal{A}^{(0,1)}[\tau, \rho] r^{(1,1)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho]) (-2 r^{(1,2)}[\tau, \rho] + \mathcal{A}^{(1,0)}[\tau, \rho] r^{(2,0)}[\tau, \rho] + \\
& \quad 2 (-1 + \mathcal{A}[\tau, \rho]) r^{(3,0)}[\tau, \rho])) + r[\tau, \rho] (\mathcal{A}^{(1,0)}[\tau, \rho]^3 - \\
& \quad 2 (-1 + \mathcal{A}[\tau, \rho]) \mathcal{A}^{(1,0)}[\tau, \rho] \mathcal{A}^{(2,0)}[\tau, \rho] + (-1 + \mathcal{A}[\tau, \rho])^2 \mathcal{A}^{(3,0)}[\tau, \rho])) = 0
\end{aligned}$$

Kinetic scalar X_E

Kinetic scalar (component form):

$$\frac{\phi^{(0,1)}[\tau, \rho]^2}{1 - \mathcal{A}[\tau, \rho]} + \phi^{(1,0)}[\tau, \rho]^2$$

Equation export